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MIL-M-28787/540
10 MAY 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

HIGH PERFORMANCE EMULATION CONTROL I, KEY CODE EWAH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic high performance emulation control I module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-5523, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.613 failures per million operating hours for class I or 8.943 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be EWAH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc Stress	IV	S	--Module shall pass all remaining tests--------						mA
D1,D2	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D3	Ibd	V	S	0.0	1.3	0.0	1.3	0.0	1.3	mA
D4	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D5	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	mA
D6,D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10-D17	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D18-D23	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D24-D27	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D28	Ibd	V	S	0.0	8.0	0.0	8.0	0.0	8.0	mA
D29-D36	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D37	Ibd	V	S	0.0	1.3	0.0	1.3	0.0	1.3	mA
D38-D40	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D41	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D42	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D43,D44	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D45	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D46	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D47,D48	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D49	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1-E17	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E18-E21	Ios	VI	S	- 85.0	-12.0	- 85.0	-12.0	- 85.0	-12.0	mA
E22	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E23-E41	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C		Units
				Min	Max	Min	Max			
F1-F4	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F5-F8	Iih	VII	X	-1.15	0.12	-1.15	0.12	-1.15	0.12	mA
F9,F10	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F11	Iih	VII	X	-2.05	0.085	-2.05	0.085	-2.05	0.085	mA
F12	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F13	Iih	VII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
F14,F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16	Iih	VII	X	-1.45	0.065	-1.45	0.065	-1.45	0.065	mA
F17	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F18-F25	Iih	VII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
F26	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F27,F28	Iih	VII	X	-1.6	0.08	-1.6	0.08	-1.6	0.08	mA
F29	Iih	VII	X	-3.2	0.16	-3.2	0.16	-3.2	0.16	mA
F30,F31	Iih	VII	X	-1.6	0.08	-1.6	0.08	-1.6	0.08	mA
F32-F35	Iih	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F36	Iih	VII	X	-16.0	0.4	-16.0	0.4	-16.0	0.4	mA
F37-F44	Iih	VII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
F45	Iih	VII	X	-2.05	0.085	-2.05	0.085	-2.05	0.085	mA
F46	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F47-F54	Iih	VII	X	-0.85	0.09	-0.85	0.09	-0.85	0.09	mA
F55-F62	Iih	VII	X	-1.05	0.11	-1.05	0.11	-1.05	0.11	mA
F63,F64	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F65	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F66	Iih	VII	X	-1.45	0.065	-1.45	0.065	-1.45	0.065	mA
F67,F68	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F69	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F70	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F71,F72	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F73	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G1-G4	III	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G5-G9	III	VIII	X	-1.15	0.12	-1.15	0.12	-1.15	0.12	mA
G9,G10	III	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G11	III	VIII	X	-2.05	0.085	-2.05	0.085	-2.05	0.085	mA
G12	III	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G13	III	VIII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
G14,G15	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16	III	VIII	X	-1.45	0.065	-1.45	0.065	-1.45	0.065	mA
G17	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G18-G25	III	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
G26	III	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G27,G28	III	VIII	X	-1.6	0.08	-1.6	0.08	-1.6	0.08	mA
G29	III	VIII	X	-3.2	0.16	-3.2	0.16	-3.2	0.16	mA
G30,G31	III	VIII	X	-1.6	0.08	-1.6	0.08	-1.6	0.08	mA
G32-G35	III	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G36	III	VIII	X	-16.0	0.4	-16.0	0.4	-16.0	0.4	mA
G37-G44	III	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
G45	III	VIII	X	-2.05	0.085	-2.05	0.085	-2.05	0.085	mA
G46	III	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G47-G54	III	VIII	X	-0.85	0.09	-0.85	0.09	-0.85	0.09	mA
G55-G62	III	VIII	X	-1.05	0.11	-1.05	0.11	-1.05	0.11	mA
G63,G64	III	VIII	X	-0.4	0.22	-0.4	0.22	-0.4	0.22	mA
G65	III	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G66	III	VIII	X	-1.45	0.065	-1.45	0.065	-1.45	0.065	mA
G67,G68	III	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G69	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G70	III	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G71,G72	III	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G73	III	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H41	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J41	Vol	X	X	0.0	0.5	0 0	0.5	0.0	0.5	V
K1-K12	Iozh	XI	X	-0.05	0.05	-0 05	0.05	-0.05	0.05	mA
K13-K24	Iozl	XI	X	-0.05	0 05	-0.05	0.05	-0.05	0.05	mA
L1,L2	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L3-L18	Vic	XII	S	-1.2	0 0	-1 2	0.0	-1.2	0.0	V
L19-L27	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L28-L37	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L38-L40	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L41-L49	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	3.2	6.1	3.2	6.1	3.2	6.1	uF
M2	PSFC	XIII	S	2.6	4.9	2.6	4.9	2.6	4.9	uF
N1	Icc	XIV	X	870	2000	870	2000	870	2000	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1,R2	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R3,R4	Ci/o	XVIII	Q	4	150	4	150	4	150	pF
R5	Ci/o	XVIII	Q	4	125	4	125	4	125	pF
R6,R7	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R8	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R9	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R10-R17	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R18	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R19,R20	Ci/o	XVIII	Q	8	100	8	100	8	100	pF
R21	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R22,R23	Ci/o	XVIII	Q	8	100	8	100	8	100	pF
R24-R27	Ci/o	XVIII	Q	2	50	2	50	2	50	pF

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R28	Ci/o	XVIII	Q	16	125	16	125	16	125	pF
R29,R30	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R31,R32	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R33-R35	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R36	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R37	Ci/o	XVIII	Q	2	175	2	175	2	175	pF
R38	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R39	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R40	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R41	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R42	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R43,R44	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R45	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R46	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R47	Ci/o	XVIII	Q	6	75	6	75	6	75	pF
R48	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R49-R53	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R54-R57	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R58	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R59-R64	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R65	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R66	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R67	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R68-R70	Ci/o	XVIII	Q	4	125	4	125	4	125	pF
R71	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R72-R74	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R75,R76	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R77	Ci/o	XVIII	Q	6	75	6	75	6	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R78-R82	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R83	Ci/o	XVIII	Q	2	150	2	150	2	150	pF
R84-R86	Ci/o	XVIII	Q	2	175	2	175	2	175	pF
R87,R88	Ci/o	XVIII	Q	2	150	2	150	2	150	pF
R89,R90	Ci/o	XVIII	Q	2	175	2	175	2	175	pF
S	Vic latch up	XIX	S	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTR	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/540-1 for class I, and M28787/540-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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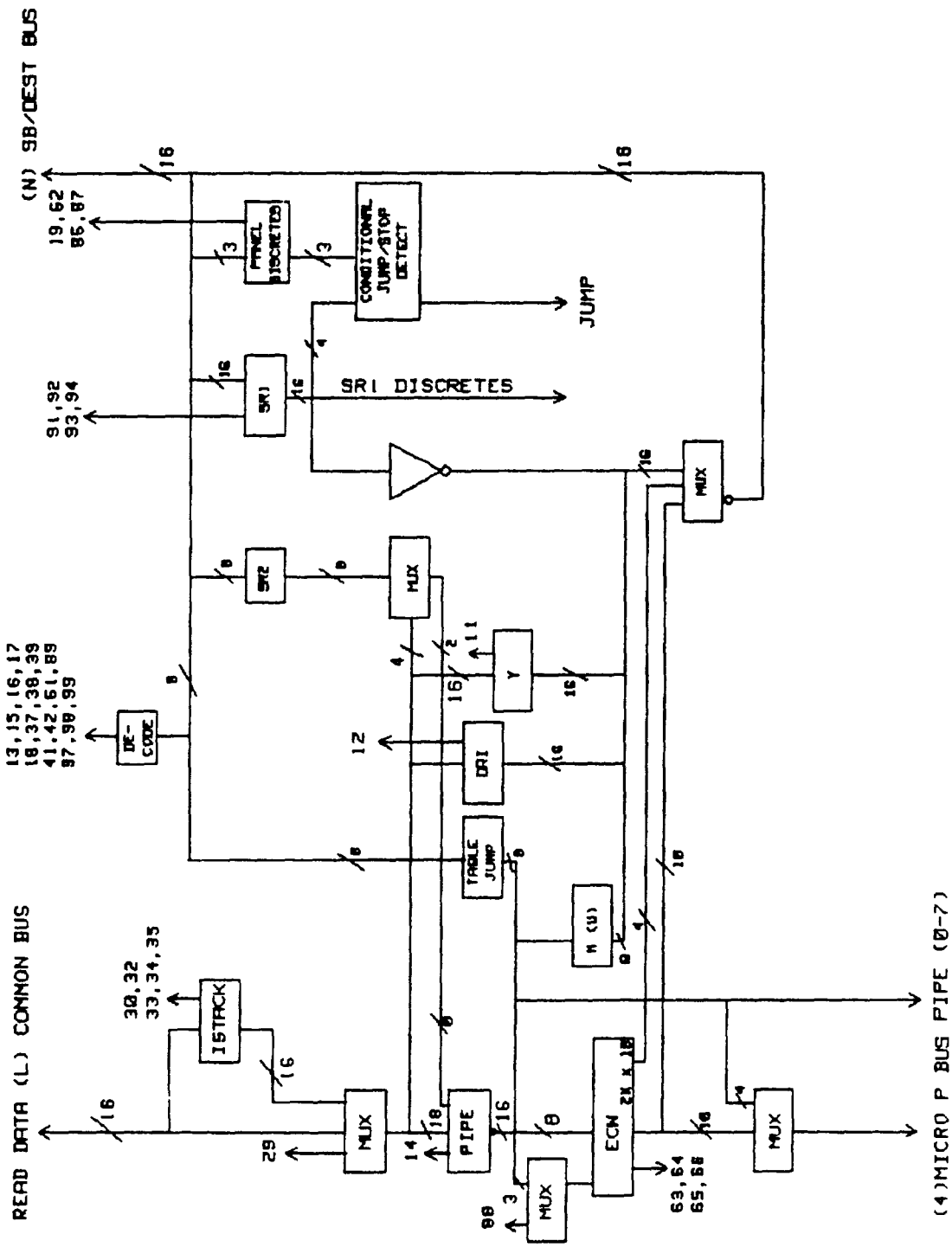


FIGURE 1. EWAH functional block diagram.

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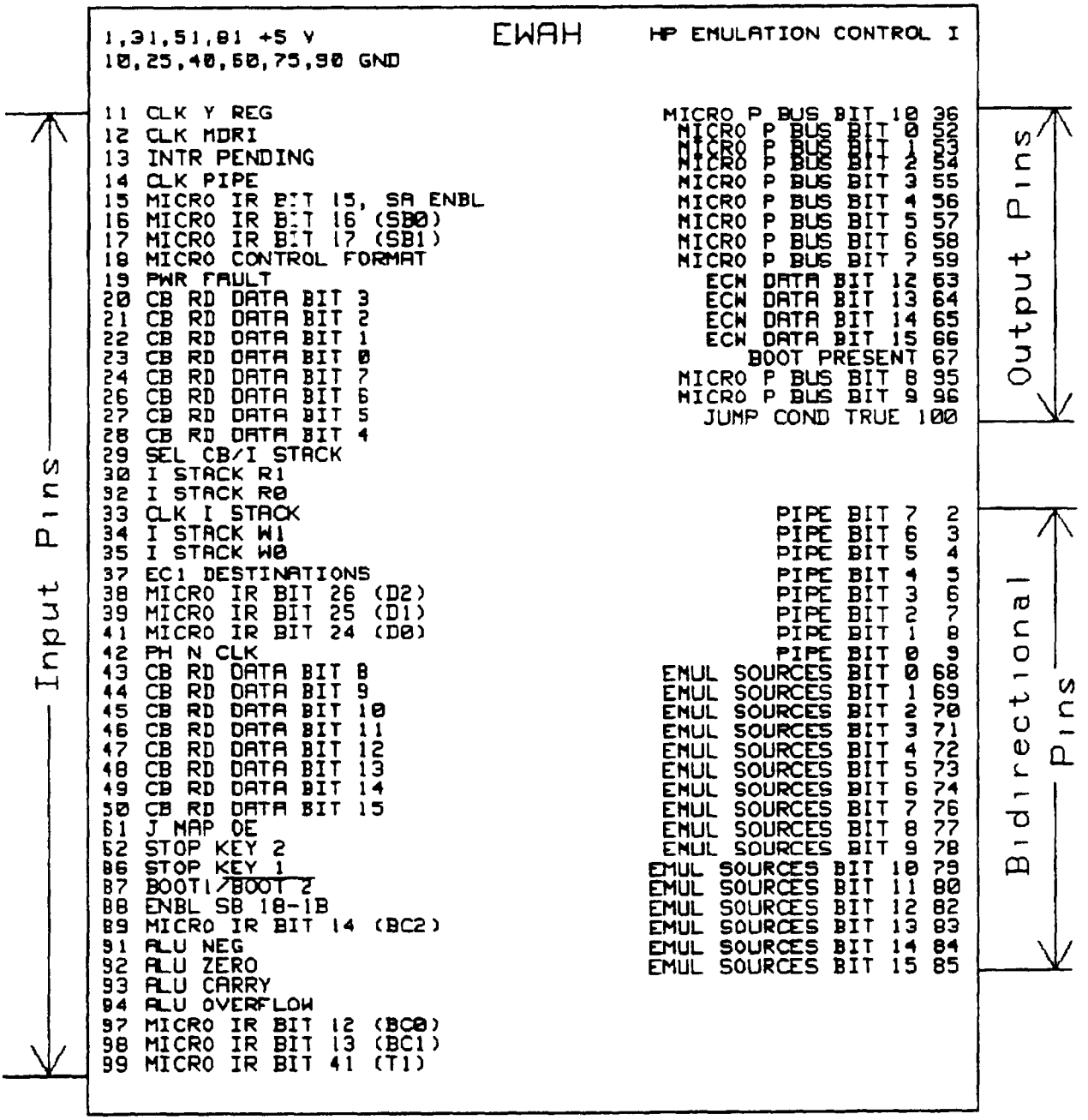


FIGURE 2. EWAH pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L and S are stress tests and will be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature

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stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

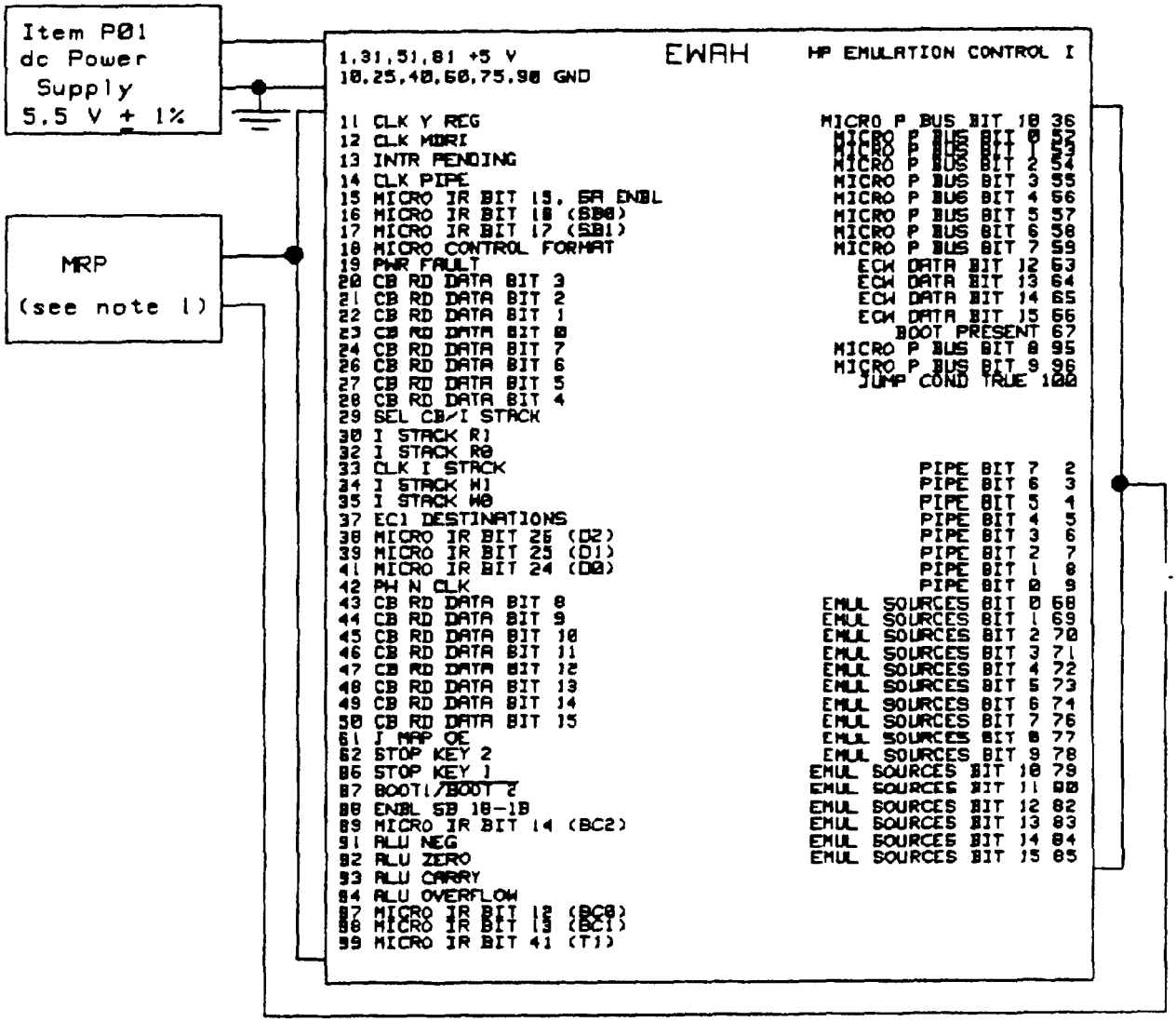
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. EWAH operating temperature and component temperature test circuit.

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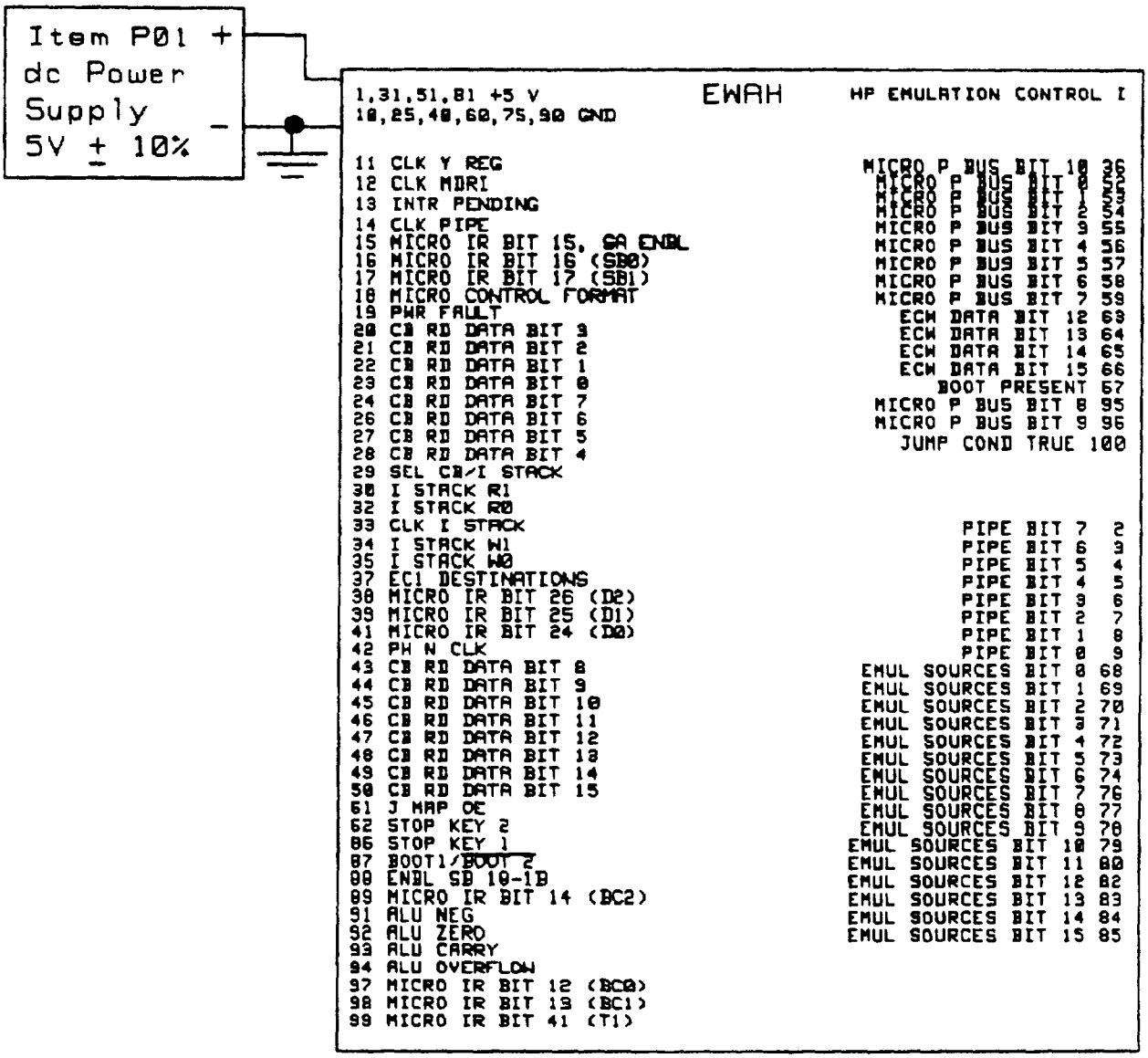
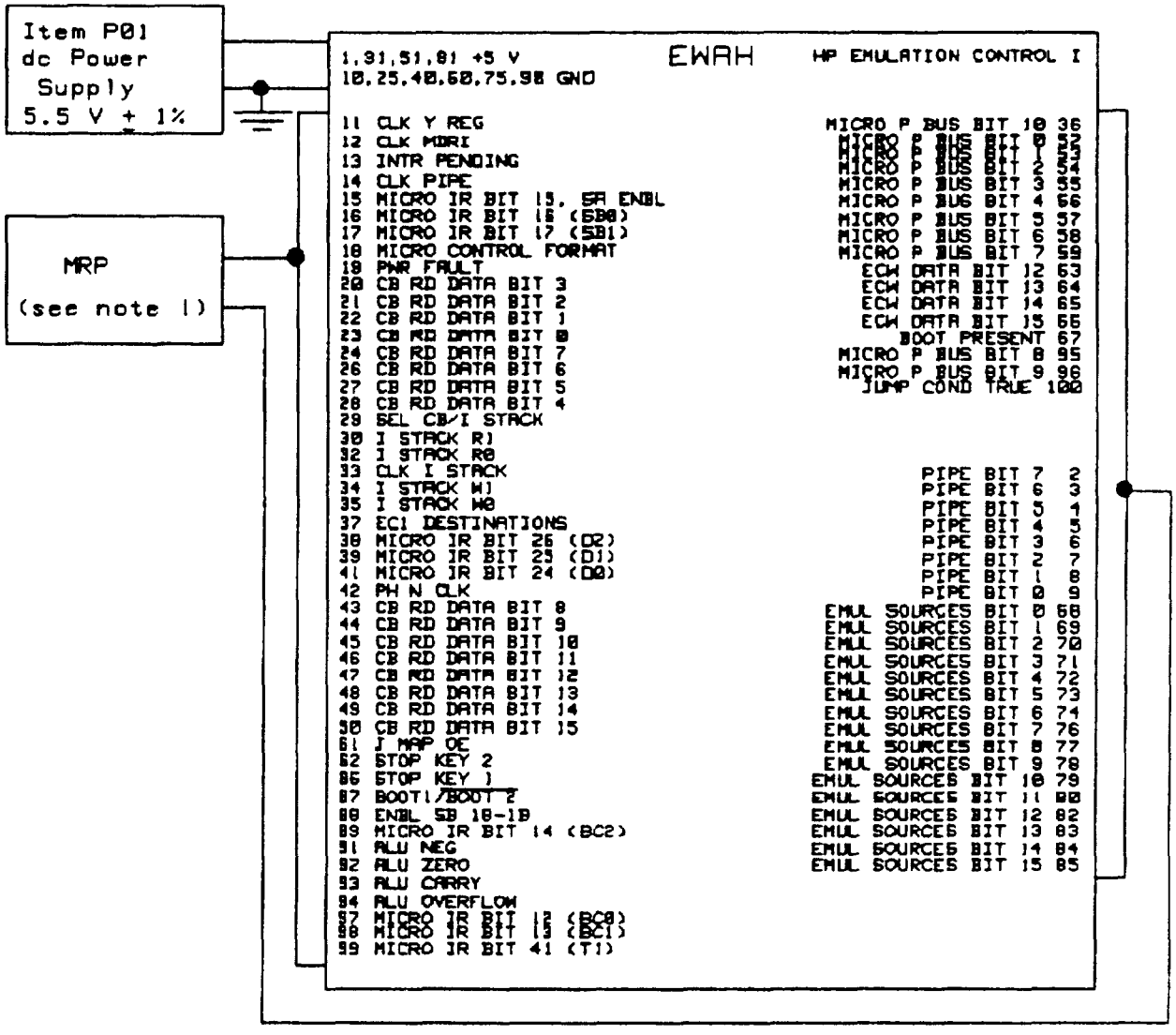


FIGURE 4. EWAH vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. EWAH life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 11.484 W (max); 7.85 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PIPE BIT 7	NA	0.4	2.4	-0.25	0.07	NA	75
PIPE BIT 6	NA	0.4	2.4	-0.25	0.07	NA	75
PIPE BIT 5	NA	0.4	2.4	-0.25	0.07	NA	75
PIPE BIT 4	NA	0.4	2.4	-0.25	0.07	NA	75
PIPE BIT 3	NA	0.4	2.4	-1.15	0.12	NA	100
PIPE BIT 2	NA	0.4	2.4	-1.15	0.12	NA	100
PIPE BIT 1	NA	0.4	2.4	-1.15	0.12	NA	100
PIPE BIT 0	NA	0.4	2.4	-1.15	0.12	NA	100
EMUL BIT 0	NA	0.4	2.4	-0.85	0.09	NA	100
EMUL BIT 1	NA	0.4	2.4	-0.85	0.09	NA	100
EMUL BIT 2	NA	0.4	2.4	-0.85	0.09	NA	75
EMUL BIT 3	NA	0.4	2.4	-0.85	0.09	NA	100
EMUL BIT 4	NA	0.4	2.4	-0.85	0.09	NA	100
EMUL BIT 5	NA	0.4	2.4	-0.85	0.09	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EMUL BIT 6	NA	0.4	2.4	-0.85	0.09	NA	100
EMUL BIT 7	NA	0.4	2.4	-0.85	0.09	NA	100
EMUL BIT 8	NA	0.4	2.4	-1.05	0.11	NA	150
EMUL BIT 9	NA	0.4	2.4	-1.05	0.11	NA	175
EMUL BIT 10	NA	0.4	2.4	-1.05	0.11	NA	175
EMUL BIT 11	NA	0.4	2.4	-1.05	0.11	NA	175
EMUL BIT 12	NA	0.4	2.4	-1.05	0.11	NA	150
EMUL BIT 13	NA	0.4	2.4	-1.05	0.11	NA	150
EMUL BIT 14	NA	0.4	2.4	-1.05	0.11	NA	175
EMUL BIT 15	NA	0.4	2.4	-1.05	0.11	NA	175
CLK Y REG	-1.5	0.4	2.4	-0.4	0.04	0.2	100
CLK MDRI	-1.5	0.4	2.4	-0.4	0.04	0.2	100
CLK PIPE	-1.2	0.4	2.4	-1.6	0.06	0.3	150
IR BIT 15	-1.2	0.4	2.4	-1.1	0.07	2.1	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 17	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CTRL FRMT	-1.2	0.4	2.4	-1.45	0.065	1.2	125
PWR FAULT	-1.2	0.4	2.4	-0.6	0.02	0.1	100
DAT BIT 3	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 2	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 1	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 0	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 7	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 6	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 5	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 4	-1.2	0.4	2.4	-1.0	0.04	0.2	75
CB/I STACK	-1.2	0.4	2.4	-2.4	0.08	0.4	125
STACK R1	-1.5	0.4	2.4	-1.6	0.08	0.4	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
STACK R0	-1.5	0.4	2.4	-1.6	0.08	0.4	100
CLK STACK	-1.5	0.4	2.4	-3.2	0.16	0.4	125
STACK W1	-1.5	0.4	2.4	-1.6	0.08	0.4	100
STACK W0	-1.5	0.4	2.4	-1.6	0.08	0.4	100
EC1 DEST	-1.5	0.4	2.4	-0.25	0.025	1.0	50
IR BIT 26	-1.5	0.4	2.4	-0.25	0.025	1.0	50
IR BIT 25	-1.5	0.4	2.4	-0.25	0.025	1.0	50
IR BIT 24	-1.5	0.4	2.4	-0.25	0.025	1.0	50
PH N CLK	-1.2	0.4	2.4	-16.0	0.4	8.0	125
J MAP OE	-1.2	0.4	2.4	-2.05	0.085	1.3	175
STOP KEY 2	-1.5	0.4	2.4	-0.4	0.02	0.1	100
STOP KEY 1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
BT 1/ BT 2	-1.5	0.4	2.4	-0.4	0.02	0.1	125
EN SB 18-1B	-1.2	0.4	2.4	-2.4	0.08	0.4	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 14	-1.2	0.4	2.4	-1.45	0.065	1.2	100
ALU NEG	-1.2	0.4	2.4	-1.2	0.04	0.2	75
ALU ZERO	-1.2	0.4	2.4	-1.2	0.04	0.2	75
ALU CARRY	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ALU OVERFLO	-1.2	0.4	2.4	-1.2	0.04	0.2	75
IR BIT 12	-1.2	0.4	2.4	-1.8	0.06	0.3	75
IR BIT 13	-1.2	0.4	2.4	-1.8	0.06	0.3	100
IR BIT 41	-1.2	0.4	2.4	-1.2	0.04	0.2	75
INTR PEND	-1.2	0.4	2.4	-2.05	0.085	1.3	150
DAT BIT 8	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 9	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 10	-1.2	0.4	2.4	-1.0	0.04	0.2	100
DAT BIT 11	-1.2	0.4	2.4	-1.0	0.04	0.2	100
DAT BIT 12	-1.2	0.4	2.4	-1.0	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DAT BIT 13	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 14	-1.2	0.4	2.4	-1.0	0.04	0.2	75
DAT BIT 15	-1.2	0.4	2.4	-1.0	0.04	0.2	100

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
PIPE BIT 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
PIPE BIT 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
PIPE BIT 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
PIPE BIT 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
PIPE BIT 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
PIPE BIT 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
PIPE BIT 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
PIPE BIT 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
EMUL BIT 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
EMUL BIT 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
EMUL BIT 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
EMUL BIT 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
EMUL BIT 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
EMUL BIT 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
EMUL BIT 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
EMUL BIT 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
EMUL BIT 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
EMUL BIT 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
BUS BIT 10	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	75
BUS BIT 0	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
BUS BIT 1	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
BUS BIT 2	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
BUS BIT 3	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
BUS BIT 4	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
BUS BIT 5	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
BUS BIT 6	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	75
BUS BIT 7	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
DAT BIT 12	2.4	0.5	-0.4	6.0	NA	NA	-12.0	100
DAT BIT 13	2.4	0.5	-0.4	6.0	NA	NA	-12.0	125
DAT BIT 14	2.4	0.5	-0.16	1.6	NA	NA	-12.0	125
DAT BIT 15	2.4	0.5	-0.16	1.6	NA	NA	-12.0	125
BOOT PRES	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
BUS BIT 8	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
BUS BIT 9	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50
JMP COND T	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	50

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	STOP KEY 2 to PH N CLK	30	30	30	--	--	--	nS	12
1	tsu	STOP KEY 1 to PH N CLK	30	30	30	--	--	--	nS	12
1	tsu	BT 1/ BT 2 to PH N CLK	30	30	30	--	--	--	nS	12
2	tpd	PH N CLK to JMP COND T	--	--	--	90	90	90	nS	12
3	tpd	PH N CLK to ECW BIT 12	--	--	--	140	140	140	nS	12
3	tpd	PH N CLK to ECW BIT 13	--	--	--	140	140	140	nS	12
3	tpd	PH N CLK to ECW BIT 14	--	--	--	140	140	140	nS	12
3	tpd	PH N CLK to ECW BIT 15	--	--	--	140	140	140	nS	12
4	tsu	ALU NEG to PH N CLK	15	15	15	--	--	--	nS	12
4	tsu	ALU CARRY to PH N CLK	15	15	15	--	--	--	nS	12
4	tsu	ALU ZERO to PH N CLK	15	15	15	--	--	--	nS	12
4	tsu	ALU OVER-FL to PH N CLK	15	15	15	--	--	--	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	EMUL BIT 0 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 1 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 2 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 3 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 4 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 5 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 6 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 7 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 8 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 9 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 10 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 11 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 12 to PH N CLK	10	10	10	--	--	--	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	EMUL BIT 13 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 14 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 15 to PH N CLK	10	10	10	--	--	--	nS	12
6	tpd	J MAP OE to BUS BIT 0	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 1	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 2	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 3	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 4	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 5	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 6	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 7	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 8	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 9	--	--	--	40	40	40	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	J MAP OE to BUS BIT 10	--	--	--	40	40	40	nS	12
7	tsu	DAT BIT 0 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 1 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 2 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 3 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 4 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 5 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 6 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 7 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 8 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 9 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 10 to CLK STACK	15	15	15	--	--	--	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	DAT BIT 11 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 12 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 13 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 14 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 15 to CLK STACK	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 0	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 1	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 2	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 3	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 4	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 5	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 6	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 7	15	15	15	--	--	--	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	CLK STACK to DAT BIT 8	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 9	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 10	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 11	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 12	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 13	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 14	15	15	15	--	--	--	nS	12
8	tsu	CLK STACK to DAT BIT 15	15	15	15	--	--	--	nS	12
9	tsu	CB/I STACK to CLK PIPE	40	40	40	--	--	--	nS	12
9	tsu	CB/I STACK to CLK Y REG	40	40	40	--	--	--	nS	12
9	tsu	CB/I STACK to CLK MDRI	40	40	40	--	--	--	nS	12
10	tsu	STACK W0 to CLK Y REG	85	85	85	--	--	--	nS	12
10	tsu	STACK W0 to CLK PIPE	85	85	85	--	--	--	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tsu	STACK W0 to CLK MDRI	85	85	85	--	--	--	nS	12
10	tsu	STACK R0 to CLK Y REG	85	85	85	--	--	--	nS	12
10	tsu	STACK R0 to CLK PIPE	85	85	85	--	--	--	nS	12
10	tsu	STACK R0 to CLK MDRI	85	85	85	--	--	--	nS	12
11	tpd	PH N CLK to PIPE BIT 0	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 1	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 2	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 3	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 4	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 5	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 6	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 7	--	--	--	50	50	50	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 0	--	--	--	25	25	25	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	EN SB 18-1B to EMUL BIT 1	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 2	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 3	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 4	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 5	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 6	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 7	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 8	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 9	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 10	--	--	--	25	25	25	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	EN SB 18-1B to EMUL BIT 11	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 12	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 13	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 14	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 15	--	--	--	25	25	25	nS	12

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6.4 Module function. Emulation control provides the control logic to emulate the macroinstructions, process interrupts, communicate over the common bus, and master clear the MRP. High performance emulation control is contained on three modules; Emulation Control 1 (EWAH), 2 (FWA), and 3 (FWB). Macroinstructions are loaded into the pipe register from memory via the common bus. For double length instructions, the Y word is loaded into the Y register. Macroinstruction emulation is performed by executing a series of short microprogram subroutines stored in micro memory. The subroutine starting address is contained in an Emulation Control Word (ECW) stored in the ECW Programmable Read Only Memory (PROM). The ECW contains control bits along with the address pointer. A group of eight words are associated with each macroinstruction. The group is addressed by the macroinstruction op and format codes (bits 15-8). An ECW within the group is addressed by address control. Address control determines if indexing, indirect addressing or both is specified, the instruction is being executed remotely, and the interim sequences required to execute the instruction. Status register 2 (bits 15-18) holds the indirect addressing information. Status register 1 holds Central Processor (CP) status bits that provide interrupt class lockouts, results of arithmetic operations, selection of page register sets, general register sets, Non-Destructive Readout (NDRO) memory, and CP mode selection (executive mode or task mode). The executive mode has a separate set of privileged instructions that cannot be executed in the task mode (an attempt to execute a privileged instruction in the task mode results in a class II interrupt). Executive mode instructions are flagged. Conditional jump instructions (Op code = 40 (base 8)) are interpreted by conditional jump/stop control to determine if the specified jump condition is true. If true, the next macroinstruction is fetched from the jump address memory location. Interrupt control detects interrupts, determines the interrupt with the highest priority and generates a 4-bit interrupt pointer. The interrupt pointer is used to form the starting address of the microprogram subroutine that processes the interrupt. Bus master control provides the CP control interface with the common bus. The CP is always the bus master when communicating with the Input/Output Controllers (IOCs) or memory interfaces. The bus is requested from the bus arbitrator and when the request is granted, bus master control generates a 4-bit function code and the control signals to perform the transaction. A MRP master clear is initiated when the MRP is powered-up or by the maintenance processor. Master clear places the MRP in the following initial condition:

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(a) P register cleared.

(b) Status registers 1 and 2 cleared.

(c) Real Time Clock (RTC) and monitor clock functions disabled.

(d) Page registers are set to their own register address.

(e) All I/O channels cleared as follows:

1. Status words cleared.
2. All data buffers and chains deactivated.
3. All interrupt data storage disabled, all External Interrupt Enable (EIE) cleared.
4. Class III interrupts disabled.
5. Serial monitor and suppress flags cleared.
6. Intercomputer time out function cleared.

(f) Bus initialization signal generated.

(g) Run mode selected (only occurs on power-up master clear).

(h) Program and power fault cleared.

6.4.1 Pipe register. The 16-bit pipe register holds the macroinstruction. The instruction is loaded from memory via the common bus and instruction multiplexer. The register output is enabled provided the microinstruction D-field is not equal to 03.

6.4.2 Y register. The 16-bit Y register holds the macroinstruction Y word. The word is loaded from memory via the common bus and instruction multiplexer. The register output is enabled by a microinstruction with SB = 19.

6.4.3 Emulation Control Word (ECW) PROM. The ECW PROM contains 2048 6-bit words. The ECWs are used to enter the correct microprogram routine for macroinstruction execution. Each instruction is assigned eight ECW words. The ECW listing is contained in the CP Microprogram Listing. The ECW PROM is addressed by a 10-bit address word (one's complement). The address is formed from the macroinstruction upper eight bits (address bits 10-3) and the ECW address 2-0 selector output. The macroinstruction 6-bit Op code and 2-bit format code address the group of eight ECWs associated with the instruction and the

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output from the selector addresses one word in the group. The ECW lower 10 bits contain the starting address of a microprogram subroutine. Bit 10 is the unary control bit. The unary macroinstructions use the m-field as a secondary operation code, thus bit 10 is used to modify the address pointer by inserting the m-field into address pointer bits 6-3. This permits starting at different microprogram addresses for each m-field value. Bit 11 enables interpretation of the indirect addressing indexing bit. If indexing is specified (determined by indirect control bits in Source Register (SR) 2), the address pointer is modified to jump to the indirect addressing with indexing subroutine. Bits 13 and 12 are memory control bits. Bits 15 and 14 are Select General Register (SGR) control bits.

6.4.4 Address control bit decoder. The address control bit decoder interprets the SR2 indirect control bits and macroinstruction m-field to determine if indirect addressing is specified. The outputs are used to point to the indirect addressing microprogram routine.

6.4.5 ECW address 2-0 selector. The selector determines the source for ECW address bits 2-0 (one's complement). When microinstruction bit 41 is a zero, the source selected is the address control bit decoder. When bit 41 is one, the source selected is microinstruction bits 14-12 (BC2-BC0).

6.4.6 Micro P selector. The 10-bit micro P selector forms the emulation microprogram starting address. The selector is enabled by an emulate microinstruction (P = 2). If an interrupt is pending, zeros are inserted in bits 9-7 and 2-0. Bits 6-3 are inserted on the micro P bus from the interrupt pointer register (emulation control 2 module). With no interrupts pending, ECW bits 11 and 10 determine how the address is formed. When ECW bits 11 and 10 are zero, ECW bits 9-0 are selected to form the address. When ECW bit 10 is a one, pipe register bits 3-0 are placed in bits 6-3 to form the address. When ECW bit 11 is a one, the indirect indexing bit from the address control bit decoder is placed in bit 9 to form the address.

6.4.7 Status register 2 upper. The 8-bit SR2 upper holds the indirect control bits. The register is loaded by a microinstruction with D = 1F.

6.4.8 Table jump register. The 8-bit register holds the jump table address. The register is loaded and enabled by a microinstruction with D = 03. The output is used as ECW

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address bits 10-3 or the very next microinstruction cycle following the load microinstruction.

6.4.9 Instruction register upper. The 8-bit register holds the macroinstruction upper bits (operation code and format code). The register is loaded by an emulate microinstruction (P = 2 and bit 41 = 0) or by a microinstruction with D = OE. The register output is enabled by a microinstruction with SB = 1B.

6.4.10 Status register 1. The 16-bit SR1 holds status B/destination bus by a microinstruction with D = 01. Bits 11-8 (carry, overflow, and condition code designators) can be loaded selectively via the SR1 11-8 selector by a format 3 microinstruction. The register output is enabled by a microinstruction with Source B (SB) = 18.

6.4.11 Emulation source selector. The 16-bit selector places the ECW word or emulation S bus on the source B/destination bus. The ECW word is selected when discrete register bit 5 = 0 and a microinstruction with SB = 18 is executed. The selector output is enabled by a microinstruction with SB = 18, 19, 1A, or 1B.

6.4.12 Discrete register. The 8-bit register holds discrete information. The register is loaded from the source B/destination bus bits 7-0 by a microinstruction with D = 02.

6.4.13 Conditional jump/stop detector. The detector determines if the Op Stop (OS) or Op Jump (OJ) condition exists. The detector is a Programmable Array Logic (PAL) device.

6.4.14 Micro branch selectors. The micro branch selectors (emulation control 1 and 2 modules) determine if the branch condition is true. The branch condition to interrogate is selected by the microinstruction BC field.

6.4.15 Destination decoder. The destination decoder translates the microinstruction D-field to provide register enables on the module.

6.4.16 Common bus and page control circuits. The common bus and page control circuits provide common bus function codes and control signals to transfer addresses and macroinstructions to/from memory via the common bus. Four PALs provide the page and function controls.

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6.4.17 Instruction fault logic. The instruction fault logic is composed of two PALs that check the macroinstructions versus the options available and the mode of operation. A CP instruction fault is generated when one of the following conditions occur:

(a) The RTC/monitor clock is not installed and an attempt is made to execute an RTC/monitor clock macroinstruction (OC, m=3, 7, 8, F, and 10, m=4, 6).

(b) The Memory Address Expansion (MAE) module is not installed and an attempt is made to execute an MAE macroinstruction (B0, B1, B3 through B5, B7, E0, E3, E7, EB, EF, F1, F3).

(c) An IOC is not installed and an attempt is made to execute the IOC command start macroinstruction (74).

An exec instruction fault is generated when an attempt is made to execute an executive mode instruction in the task mode. The CP instruction fault output from the instruction fault PAL is sent to the micro-P selector to be placed onto the micro-P bus. Exec fault signal is sent to the micro-P bus selector to be used as a selector enable. Micro-P bus bits 6-3 are used to form the emulation program starting address.

6.4.18 Source decoder. The source decoder decodes the least two significant bits of the microinstruction SB field. Outputs from the source decoder are source enabled to register on the emulation source bus.

6.4.19 Instruction stack. The instruction stack (ISTACK) is a 4-word by 16-bit file used to hold macroinstructions read up from main memory via the common bus (CB Read Data 15-0). ISTACK control logic is contained on the emulation control 2 and 3 modules.

6.4.20 Memory Data Register Input (MDRI) register. The MDRI register is used to hold single word operand reads and the first word of a double word read. The MDRI is clocked by the MC field decoder PAL on the emulation control 3 module and is enabled onto the emulation source bus by the microinstruction SB field (bits 17, 16 equal to 1, 0).

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6.4.21 Description of I/O pins.

Signal name	Pin number	I/O	Description
PIPE BIT 7	2	I/O	PIPE BIT 7
PIPE BIT 6	3	I/O	PIPE BIT 6
PIPE BIT 5	4	I/O	PIPE BIT 5
PIPE BIT 4	5	I/O	PIPE BIT 4
PIPE BIT 3	6	I/O	PIPE BIT 3
PIPE BIT 2	7	I/O	PIPE BIT 2
PIPE BIT 1	8	I/O	PIPE BIT 1
PIPE BIT 0	9	I/O	PIPE BIT 0
EMUL BIT 0	68	I/O	EMULATION SOURCES BIT 0
EMUL BIT 1	69	I/O	EMULATION SOURCES BIT 1
EMUL BIT 2	70	I/O	EMULATION SOURCES BIT 2
EMUL BIT 3	71	I/O	EMULATION SOURCES BIT 3
EMUL BIT 4	72	I/O	EMULATION SOURCES BIT 4
EMUL BIT 5	73	I/O	EMULATION SOURCES BIT 5
EMUL BIT 6	74	I/O	EMULATION SOURCES BIT 6
EMUL BIT 7	76	I/O	EMULATION SOURCES BIT 7
EMUL BIT 8	77	I/O	EMULATION SOURCES BIT 8
EMUL BIT 9	78	I/O	EMULATION SOURCES BIT 9
EMUL BIT 10	79	I/O	EMULATION SOURCES BIT 10
EMUL BIT 11	80	I/O	EMULATION SOURCES BIT 11

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6.4.21 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EMUL BIT 12	82	I/O	EMULATION SOURCES BIT 12
EMUL BIT 13	83	I/O	EMULATION SOURCES BIT 13
EMUL BIT 14	84	I/O	EMULATION SOURCES BIT 14
EMUL BIT 15	85	I/O	EMULATION SOURCES BIT 15
CLK Y REG	11	I	CLK Y REG
CLK MDRI	12	I	CLK MDRI
INTR PEND	13	I	INTR PEND
CLK PIPE	14	I	CLK PIPE
IR BIT 15	15	I	MICRO IR BIT 15, SA ENBL
IR BIT 16	16	I	MICRO IR BIT 16, (SB0)
IR BIT 17	17	I	MICRO IR BIT 17, (SB1)
CTRL FRMT	18	I	MICRO CONTROL FORMAT
PWR FAULT	19	I	PWR FAULT
DAT BIT 3	20	I	CB RD DATA BIT 3
DAT BIT 2	21	I	CB RD DATA BIT 2
DAT BIT 1	22	I	CB RD DATA BIT 1
DAT BIT 0	23	I	CB RD DATA BIT 0
DAT BIT 7	24	I	CB RD DATA BIT 7
DAT BIT 6	26	I	CB RD DATA BIT 6
DAT BIT 5	27	I	CB RD DATA BIT 5
DAT BIT 4	28	I	CB RD DATA BIT 4

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6.4.21 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB/I STACK	29	I	SEL CB/I STACK
STACK R1	30	I	I STACK R1
STACK R0	32	I	I STACK R0
CLK STACK	33	I	CLK I STACK
STACK W1	34	I	I STACK W1
STACK W0	35	I	I STACK W0
EC1 DEST	37	I	EC1 DESTINATIONS
IR BIT 26	38	I	MICRO IR BIT 26 (D2)
IR BIT 25	39	I	MICRO IR BIT 25 (D1)
IR BIT 24	41	I	MICRO IR BIT 24 (D0)
PH N CLK	42	I	PH N CLK
DAT BIT 8	43	I	CB RD DATA BIT 8
DAT BIT 9	44	I	CB RD DATA BIT 9
DAT BIT 10	45	I	CB RD DATA BIT 10
DAT BIT 11	46	I	CB RD DATA BIT 11
DAT BIT 12	47	I	CB RD DATA BIT 12
DAT BIT 13	48	I	CB RD DATA BIT 13
DAT BIT 14	49	I	CB RD DATA BIT 14
DAT BIT 15	50	I	CB RD DATA BIT 15
J MAP OE	61	I	J MAP OE
STOP KEY 2	62	I	STOP KEY 2

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6.4.21 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
STOP KEY 1	86	I	STOP KEY 1
BT 1/ BT 2	87	I	BOOT 1/ BOOT 2
EN SB 18-1B	88	I	EN SB 18-1B
IR BIT 14	89	I	MICRO IR BIT 14 (BC2)
ALU NEG	91	I	ALU NEG
ALU ZERO	92	I	ALU ZERO
ALU CARRY	93	I	ALU CARRY
ALU OVERFL	94	I	ALU OVERFLOW
IR BIT 12	97	I	MICRO IR BIT 12 (BC0)
IR BIT 13	98	I	MICRO IR BIT 13 (BC1)
IR BIT 41	99	I	MICRO IR BIT 41 (T1)
BUS BIT 10	36	O	MICRO P BUS BIT 10
BUS BIT 0	52	O	MICRO P BUS BIT 0
BUS BIT 1	53	O	MICRO P BUS BIT 1
BUS BIT 2	54	O	MICRO P BUS BIT 2
BUS BIT 3	55	O	MICRO P BUS BIT 3
BUS BIT 4	56	O	MICRO P BUS BIT 4
BUS BIT 5	57	O	MICRO P BUS BIT 5
BUS BIT 6	58	O	MICRO P BUS BIT 6
BUS BIT 7	59	O	MICRO P BUS BIT 7
DATA BIT 12	63	O	ECW DATA BIT 12

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6.4.21 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DATA BIT 13	64	O	ECW DATA BIT 13
DATA BIT 14	65	O	ECW DATA BIT 14
DATA BIT 15	66	O	ECW DATA BIT 15
BOOT PRES	67	O	BOOT PRESENT
BUS BIT 8	95	O	MICRO P BUS BIT 8
BUS BIT 9	96	O	MICRO P BUS BIT 9
JMP COND T	100	O	JUMP CONDITION TRUE

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N711)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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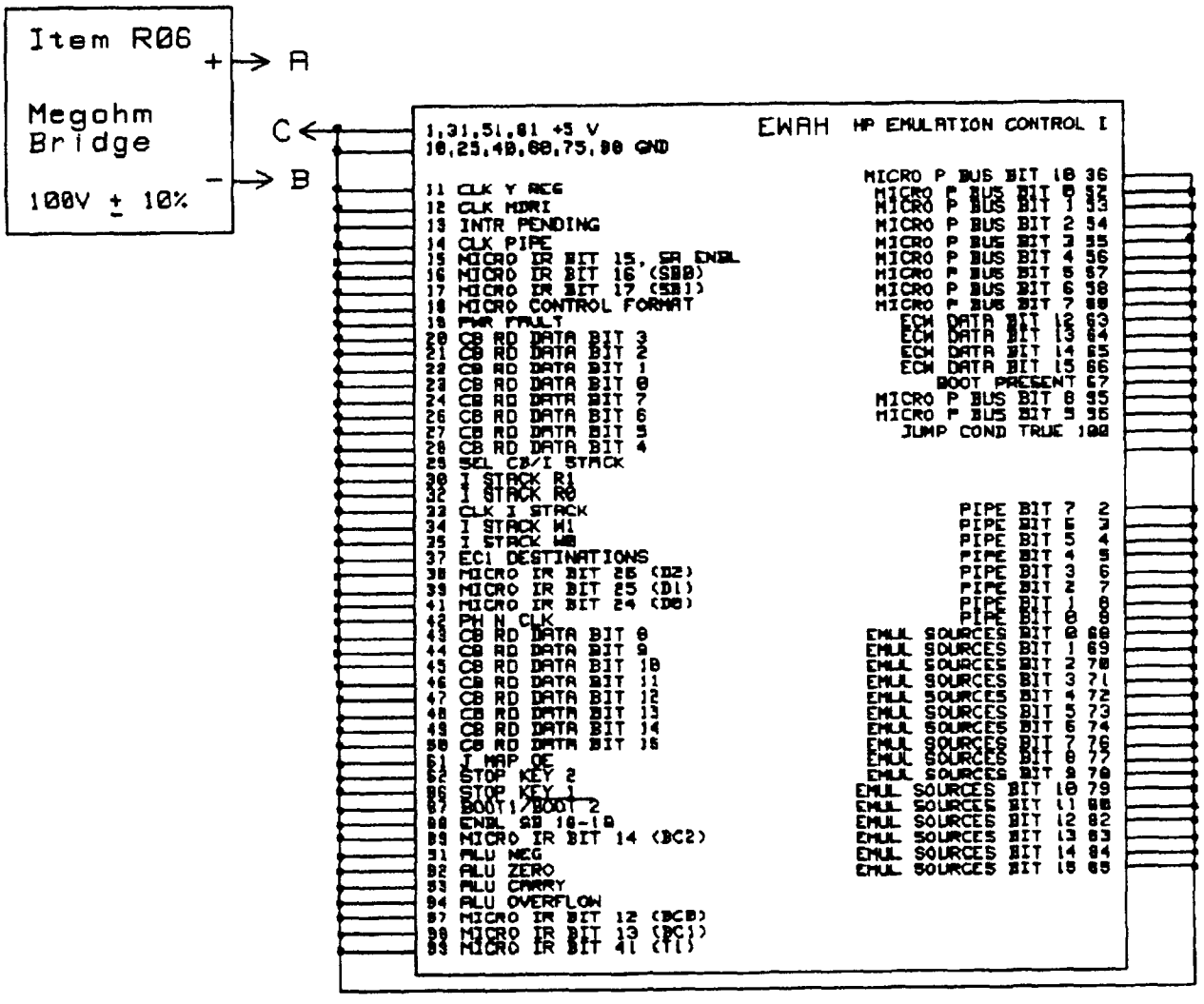


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40	0	1	Ohm	1-5
B3	Continuity, GND	60	75	0	1	Ohm	1-5
B4	Continuity, GND	60	90	0	1	Ohm	1-5
B5	Continuity, Vcc	1	31	0	1	Ohm	1-5
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Continuity measurement	NA	± 0.05 Ohm	NA	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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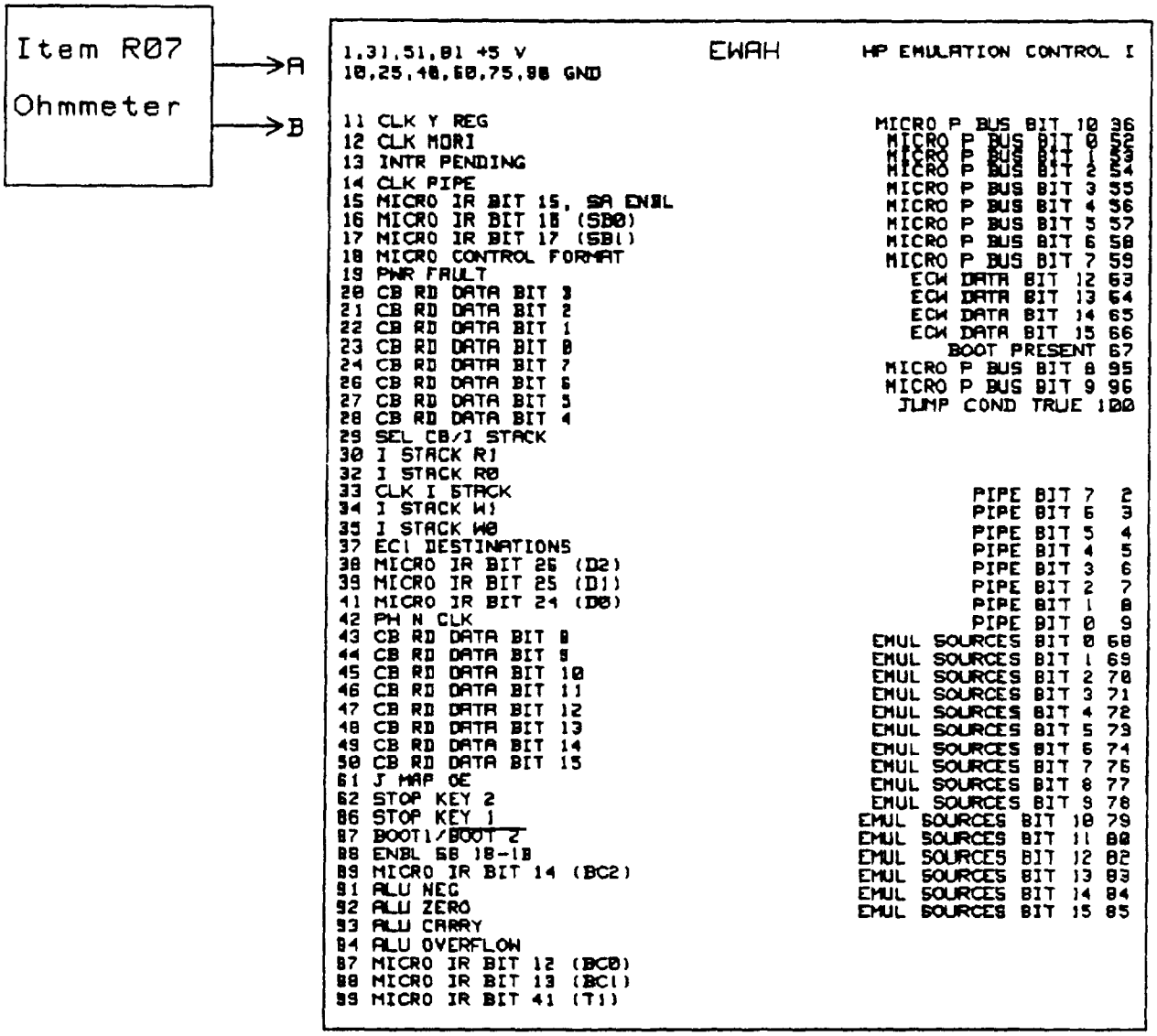


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.76 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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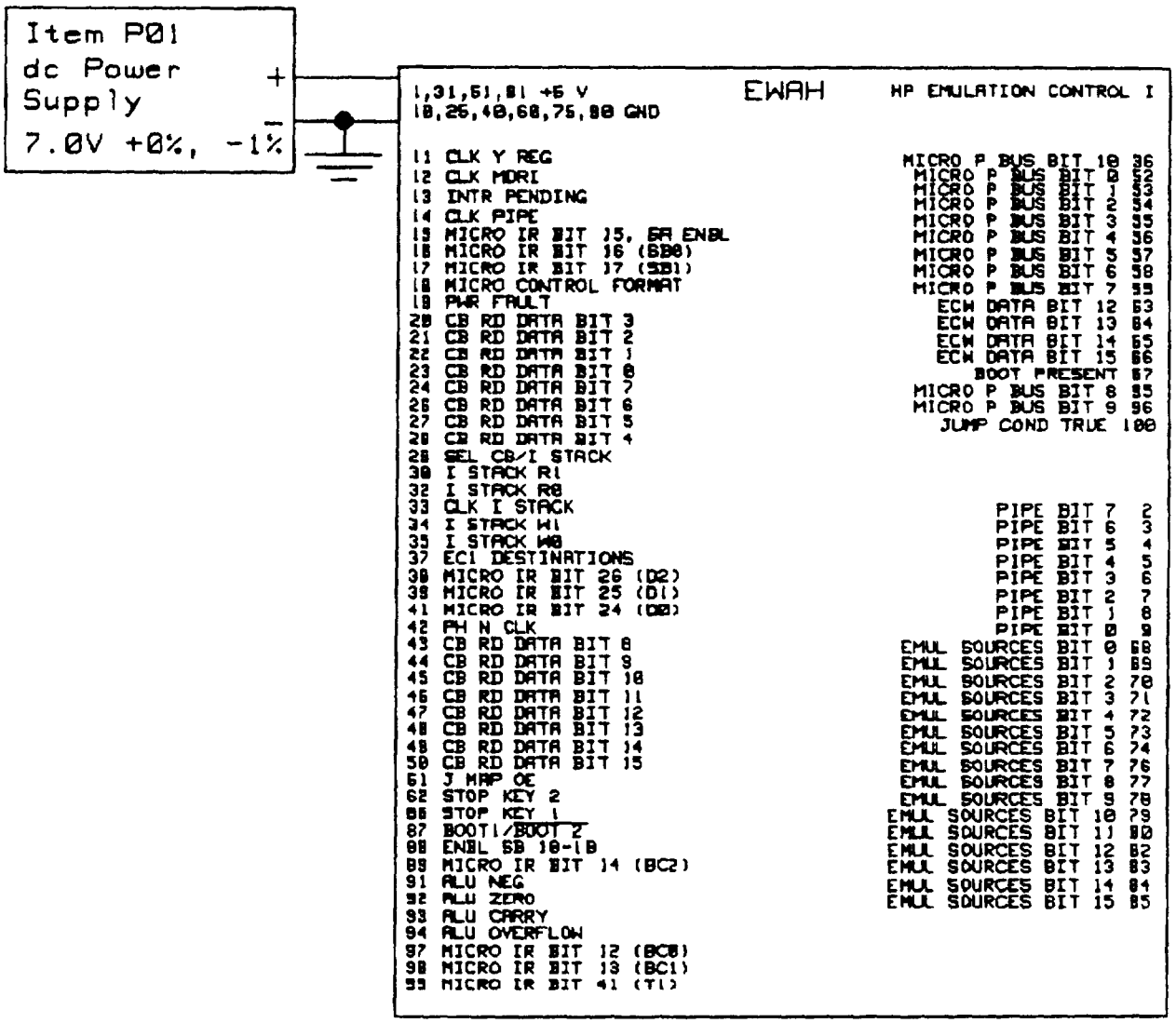


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CLK Y REG	11	5.5	0.0	0.2	mA	1-3
D2	CLK MDRI	12	5.5	0.0	0.2	mA	1-3
D3	INTR PEND	13	5.5	0.0	1.3	mA	1-3
D4	CLK PIPE	14	5.5	0.0	0.3	mA	1-3
D5	IR BIT 15	15	5.5	0.0	2.1	mA	1-3
D6	IR BIT 16	16	5.5	0.0	0.1	mA	1-3
D7	IR BIT 17	17	5.5	0.0	0.1	mA	1-3
D8	CTRL FRMT	18	5.5	0.0	1.2	mA	1-3
D9	PWR FAULT	19	5.5	0.0	0.1	mA	1-3
D10	DAT BIT 3	20	5.5	0.0	0.2	mA	1-3
D11	DAT BIT 2	21					
D12	DAT BIT 1	22					
D13	DAT BIT 0	23					
D14	DAT BIT 7	24					
D15	DAT BIT 6	26					
D16	DAT BIT 5	27					
D17	DAT BIT 4	28	5.5	0.0	0.2	mA	1-3
D18	CB/I STACK	29	5.5	0.0	0.4	mA	1-3
D19	STACK R1	30					
D20	STACK R0	32					
D21	CLK STACK	33	5.5	0.0	0.4	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	STACK W1	34	5.5	0.0	0.4	mA	1-3
D23	STACK W0	35	5.5	0.0	0.4	mA	1-3
D24	EC1 DEST	37	5.5	0.0	1.0	mA	1-3
D25	IR BIT 26	38					
D26	IR BIT 25	39					
D27	IR BIT 24	41	5.5	0.0	1.0	mA	1-3
D28	PH N CLK	42	5.5	0.0	8.0	mA	1-3
D29	DAT BIT 8	43	5.5	0.0	0.2	mA	1-3
D30	DAT BIT 9	44					
D31	DAT BIT 10	45					
D32	DAT BIT 11	46					
D33	DAT BIT 12	47					
D34	DAT BIT 13	48					
D35	DAT BIT 14	49					
D36	DAT BIT 15	50	5.5	0.0	0.2	mA	1-3
D37	J MAP OE	61	5.5	0.0	1.3	mA	1-3
D38	STOP KEY 2	62	5.5	0.0	0.1	mA	1-3
D39	STOP KEY 1	86					
D40	BT 1/ BT 2	87	5.5	0.0	0.1	mA	1-3
D41	EN SB 18-1B	88	5.5	0.0	0.4	mA	1-3
D42	IR BIT 14	89	5.5	0.0	1.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D43	ALU NEG	91	5.5	0.0	0.2	mA	1-3
D44	ALU ZERO	92	5.5	0.0	0.2	mA	1-3
D45	ALU CARRY	93	5.5	0.0	0.1	mA	1-3
D46	ALU OVERFL	94	5.5	0.0	0.2	mA	1-3
D47	IR BIT 12	97	5.5	0.0	0.3	mA	1-3
D48	IR BIT 13	98	5.5	0.0	0.3	mA	1-3
D49	IR BIT 41	99	5.5	0.0	0.2	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	3.13 A	D1-D49
Forcing voltage, (Vin)	5.5 V	± 50 mV	NA	D1-D49
Pattern voltage				
high, (VH)	3.0 V	± 100 mV	NA	D1-D49
low, (VL)	0.0 V	± 100 mV	NA	D1-D49
Input current measurement	NA	± 10 uA	NA	D1,D2
	NA	± 65 uA	NA	D3
	NA	± 15 uA	NA	D4
	NA	± 105 uA	NA	D5
	NA	± 5 uA	NA	D6,D7

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Input current measurement	NA	± 60 uA	NA	D8
	NA	± 5 uA	NA	D9
	NA	± 10 uA	NA	D10-D17
	NA	± 20 uA	NA	D18-D23
	NA	± 50 uA	NA	D24-D27
	NA	± 400 uA	NA	D28
	NA	± 10 uA	NA	D29-D36
	NA	± 65 uA	NA	D37
	NA	± 5 uA	NA	D38-D40
	NA	± 20 uA	NA	D41
	NA	± 60 uA	NA	D42
	NA	± 10 uA	NA	D43,D44
	NA	± 5 uA	NA	D45
	NA	± 10 uA	NA	D46
	NA	± 15 uA	NA	D47,D48
	NA	± 10 uA	NA	D49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	PIPE BIT 7	2	0.0	-150.0	-60.0	mA	1-4
E2	PIPE BIT 6	3					
E3	PIPE BIT 5	4					
E4	PIPE BIT 4	5					
E5	PIPE BIT 3	6					
E6	PIPE BIT 2	7					
E7	PIPE BIT 1	8					
E8	PIPE BIT 0	9					
E9	BUS BIT 10	36					
E10	BUS BIT 0	52					
E11	BUS BIT 1	53					
E12	BUS BIT 2	54					
E13	BUS BIT 3	55					
E14	BUS BIT 4	56					
E15	BUS BIT 5	57					
E16	BUS BIT 6	58					
E17	BUS BIT 7	59	0.0	-150.0	-60.0	mA	1-4
E18	DATA BIT 12	63	0.0	-85.0	-12.0	mA	1-4
E19	DATA BIT 13	64					
E20	DATA BIT 14	65					
E21	DATA BIT 15	66	0.0	-85.0	-12.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	BOOT PRES	67	0.0	-130.0	-30.0	mA	1-4
E23	EMUL BIT 0	68	0.0	-150.0	-60.0	mA	1-4
E24	EMUL BIT 1	69					
E25	EMUL BIT 2	70					
E26	EMUL BIT 3	71					
E27	EMUL BIT 4	72					
E28	EMUL BIT 5	73					
E29	EMUL BIT 6	74					
E30	EMUL BIT 7	76					
E31	EMUL BIT 8	77					
E32	EMUL BIT 9	78					
E33	EMUL BIT 10	79					
E34	EMUL BIT 11	80					
E35	EMUL BIT 12	82					
E36	EMUL BIT 13	83					
E37	EMUL BIT 14	84					
E38	EMUL BIT 15	85					
E39	BUS BIT 8	95					
E40	BUS BIT 9	96					
E41	JMP COND T	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	3.13 A	E1-E41
Forcing voltage, (Vout)	0.0 V	± 50 mV	NA	E1-E41
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 NA NA	 E1-E41 E1-E41
Output current measurement	NA NA NA NA	± 4.5 mA ± 3.65 mA ± 5.0 mA ± 4.5 mA	NA NA NA NA	E1-E17 E18-E21 E22 E23-E41

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

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NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	PIPE BIT 7	2	2.4	-0.25	0.07	mA	1-3
F2	PIPE BIT 6	3					
F3	PIPE BIT 5	4					
F4	PIPE BIT 4	5	2.4	-0.25	0.07	mA	1-3
F5	PIPE BIT 3	6	2.4	-1.15	0.12	mA	1-3
F6	PIPE BIT 2	7					
F7	PIPE BIT 1	8					
F8	PIPE BIT 0	9	2.4	-1.15	0.12	mA	1-3
F9	CLK Y REG	11	2.4	-0.4	0.04	mA	1-3
F10	CLK MDRI	12	2.4	-0.4	0.04	mA	1-3
F11	INTR PEND	13	2.4	-2.05	0.085	mA	1-3
F12	CLK PIPE	14	2.4	-1.8	0.06	mA	1-3
F13	IR BIT 15	15	2.4	-1.1	0.07	mA	1-3
F14	IR BIT 16	16	2.4	-0.6	0.02	mA	1-3
F15	IR BIT 17	17	2.4	-0.6	0.02	mA	1-3
F16	CTRL FRMT	18	2.4	-1.45	0.065	mA	1-3
F17	PWR FAULT	19	2.4	-0.6	0.02	mA	1-3
F18	DAT BIT 3	20	2.4	-1.0	0.04	mA	1-3
F19	DAT BIT 2	21					
F20	DAT BIT 1	22					
F21	DAT BIT 0	23	2.4	-1.0	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	DAT BIT 7	24	2.4	-1.0	0.04	mA	1-3
F23	DAT BIT 6	26					
F24	DAT BIT 5	27					
F25	DAT BIT 4	28	2.4	-1.0	0.04	mA	1-3
F26	CB/I STACK	29	2.4	-2.4	0.08	mA	1-3
F27	STACK R1	30	2.4	-1.6	0.08	mA	1-3
F28	STACK R0	32	2.4	-1.6	0.08	mA	1-3
F29	CLK STACK	33	0.0	-3.2	0.16	mA	1-3
F30	STACK W1	34	2.4	-1.6	0.08	mA	1-3
F31	STACK W0	35	2.4	-1.6	0.08	mA	1-3
F32	EC1 DEST	37	2.4	-0.25	0.025	mA	1-3
F33	IR BIT 26	38					
F34	IR BIT 25	39					
F35	IR BIT 24	41	2.4	-0.25	0.025	mA	1-3
F36	PH N CLK	42	2.4	-16.0	0.4	mA	1-3
F37	DAT BIT 8	43	2.4	-1.0	0.04	mA	1-3
F38	DAT BIT 9	44					
F39	DAT BIT 10	45					
F40	DAT BIT 11	46					
F41	DAT BIT 12	47					
F42	DAT BIT 13	48	2.4	-1.0	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F43	DAT BIT 14	49	2.4	-1.0	0.04	mA	1-3
F44	DAT BIT 15	50	2.4	-1.0	0.04	mA	1-3
F45	J MAP OE	61	2.4	-2.05	0.085	mA	1-3
F46	STOP KEY 2	62	2.4	-0.4	0.02	mA	1-3
F47	EMUL BIT 0	68	2.4	-0.85	0.09	mA	1-3
F48	EMUL BIT 1	69					
F49	EMUL BIT 2	70					
F50	EMUL BIT 3	71					
F51	EMUL BIT 4	72					
F52	EMUL BIT 5	73					
F53	EMUL BIT 6	74					
F54	EMUL BIT 7	76	2.4	-0.85	0.09	mA	1-3
F55	EMUL BIT 8	77	2.4	-1.05	0.11	mA	1-3
F56	EMUL BIT 9	78					
F57	EMUL BIT 10	79					
F58	EMUL BIT 11	80					
F59	EMUL BIT 12	82					
F60	EMUL BIT 13	83					
F61	EMUL BIT 14	84					
F62	EMUL BIT 15	85	2.4	-1.05	0.11	mA	1-3
F63	STOP KEY 1	86	2.4	-0.4	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F64	BT 1/ BT 2	87	2.4	-0.4	0.02	mA	1-3
F65	EN SB 18-1B	88	2.4	-2.4	0.08	mA	1-3
F66	IR BIT 14	89	2.4	-1.45	0.065	mA	1-3
F67	ALU NEG	91	2.4	-1.2	0.04	mA	1-3
F68	ALU ZERO	92	2.4	-1.2	0.04	mA	1-3
F69	ALU CARRY	93	2.4	-0.6	0.02	mA	1-3
F70	ALU OVERFL	94	2.4	-1.2	0.04	mA	1-3
F71	IR BIT 12	97	2.4	-1.8	0.06	mA	1-3
F72	IR BIT 13	98	2.4	-1.8	0.06	mA	1-3
F73	IR BIT 41	99	2.4	-1.2	0.04	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	3.13 A	F1-F73
Forcing voltage, (Vih)	2.4 V	± 50 mV	NA	F1-F73
Pattern voltage high, (VH)	3.0 V	± 100 mV	NA	F1-F73
low, (VL)	0.0 V	± 100 mV	NA	F1-F73

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Input current measurement	NA	± 16.0 uA	NA	F1-F4
	NA	± 63.5 uA	NA	F5-F8
	NA	± 22.0 uA	NA	F9,F10
	NA	± 107.0 uA	NA	F11
	NA	± 93.0 uA	NA	F12
	NA	± 58.5 uA	NA	F13
	NA	± 31.0 uA	NA	F14,F15
	NA	± 75.8 uA	NA	F16
	NA	± 31.0 uA	NA	F17
	NA	± 52.0 uA	NA	F18-F25
	NA	± 124.0 uA	NA	F26
	NA	± 84.0 uA	NA	F27,F28
	NA	± 168.0 uA	NA	F29
	NA	± 84.0 uA	NA	F30,F31
	NA	± 13.8 uA	NA	F32-F35
	NA	± 820.0 uA	NA	F36
	NA	± 52.0 uA	NA	F37-F44
	NA	± 107.0 uA	NA	F45
	NA	± 21.0 uA	NA	F46
	NA	± 47.0 uA	NA	F47-F54
	NA	± 58.0 uA	NA	F55-F62
	NA	± 21.0 uA	NA	F63,F64
	NA	± 124.0 uA	NA	F65
	NA	± 75.8 uA	NA	F66
	NA	± 62.0 uA	NA	F67,F68
	NA	± 31.0 uA	NA	F69
	NA	± 62.0 uA	NA	F70
	NA	± 93.0 uA	NA	F71,F72
	NA	± 62.0 uA	NA	F73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	PIPE BIT 7	2	0.4	-0.25	0.07	mA	1-3
G2	PIPE BIT 6	3					
G3	PIPE BIT 5	4					
G4	PIPE BIT 4	5	0.4	-0.25	0.07	mA	1-3
G5	PIPE BIT 3	6	0.4	-1.15	0.12	mA	1-3
G6	PIPE BIT 2	7					
G7	PIPE BIT 1	8					
G8	PIPE BIT 0	9	0.4	-1.15	0.12	mA	1-3
G9	CLK Y REG	11	0.4	-0.4	0.04	mA	1-3
G10	CLK MDRI	12	0.4	-0.4	0.04	mA	1-3
G11	INTR PEND	13	0.4	-2.05	0.085	mA	1-3
G12	CLK PIPE	14	0.4	-1.8	0.06	mA	1-3
G13	IR BIT 15	15	0.4	-1.1	0.07	mA	1-3
G14	IR BIT 16	16	0.4	-0.6	0.02	mA	1-3
G15	IR BIT 17	17	0.4	-0.6	0.02	mA	1-3
G16	CTRL FRMT	18	0.4	-1.45	0.065	mA	1-3
G17	PWR FAULT	19	0.4	-0.6	0.02	mA	1-3
G18	DAT BIT 3	20	0.4	-1.0	0.04	mA	1-3
G19	DAT BIT 2	21					
G20	DAT BIT 1	22					
G21	DAT BIT 0	23	0.4	-1.0	0.04	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	DAT BIT 7	24	0.4	-1.0	0.04	mA	1-3
G23	DAT BIT 6	26					
G24	DAT BIT 5	27					
G25	DAT BIT 4	28	0.4	-1.0	0.04	mA	1-3
G26	CB/I STACK	29	0.4	-2.4	0.08	mA	1-3
G27	STACK R1	30	0.4	-1.6	0.08	mA	1-3
G28	STACK R0	32	0.4	-1.6	0.08	mA	1-3
G29	CLK STACK	33	0.4	-3.2	0.16	mA	1-3
G30	STACK W1	34	0.4	-1.6	0.08	mA	1-3
G31	STACK W0	35	0.4	-1.6	0.08	mA	1-3
G32	EC1 DEST	37	0.4	-0.25	0.025	mA	1-3
G33	IR BIT 26	38					
G34	IR BIT 25	39					
G35	IR BIT 24	41	0.4	-0.25	0.025	mA	1-3
G36	PH N CLK	42	0.4	-16.0	0.4	mA	1-3
G37	DAT BIT 8	43	0.4	-1.0	0.04	mA	1-3
G38	DAT BIT 9	44					
G39	DAT BIT 10	45					
G40	DAT BIT 11	46					
G41	DAT BIT 12	47					
G42	DAT BIT 13	48	0.4	-1.0	0.04	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	DAT BIT 14	49	0.4	-1.0	0.04	mA	1-3
G44	DAT BIT 15	50	0.4	-1.0	0.04	mA	1-3
G45	J MAP OE	61	0.4	-2.05	0.085	mA	1-3
G46	STOP KEY 2	62	0.4	-0.4	0.02	mA	1-3
G47	EMUL BIT 0	68	0.4	-0.85	0.09	mA	1-3
G48	EMUL BIT 1	69					
G49	EMUL BIT 2	70					
G50	EMUL BIT 3	71					
G51	EMUL BIT 4	72					
G52	EMUL BIT 5	73					
G53	EMUL BIT 6	74					
G54	EMUL BIT 7	76	0.4	-0.85	0.09	mA	1-3
G55	EMUL BIT 8	77	0.4	-1.05	0.11	mA	1-3
G56	EMUL BIT 9	78					
G57	EMUL BIT 10	79					
G58	EMUL BIT 11	80					
G59	EMUL BIT 12	82					
G60	EMUL BIT 13	83					
G61	EMUL BIT 14	84					
G62	EMUL BIT 15	85	0.4	-1.05	0.11	mA	1-3
G63	STOP KEY 1	86	0.4	-0.4	0.22	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G64	BT 1/ BT 2	87	0.4	-0.4	0.22	mA	1-3
G65	EN SB 18-1B	88	0.4	-2.4	0.08	mA	1-3
G66	IR BIT 14	89	0.4	-1.45	0.065	mA	1-3
G67	ALU NEG	91	0.4	-1.2	0.04	mA	1-3
G68	ALU ZERO	92	0.4	-1.2	0.04	mA	1-3
G69	ALU CARRY	93	0.4	-0.6	0.02	mA	1-3
G70	ALU OVERFL	94	0.4	-1.2	0.04	mA	1-3
G71	IR BIT 12	97	0.4	-1.8	0.06	mA	1-3
G72	IR BIT 13	98	0.4	-1.8	0.06	mA	1-3
G73	IR BIT 41	99	0.4	-1.2	0.04	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	3.13 A	G1-G73
Forcing voltage, (Vil)	0.4 V	± 50 mV	NA	G1-G73
Pattern voltage, high, (VH)	3.0 V	± 100 mV	NA	G1-G73
low, (VL)	0.0 V	± 100 mV	NA	G1-G73

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Input current measurement	NA	± 16.0 uA	NA	G1-G4
	NA	± 63.5 uA	NA	G5-G8
	NA	± 22.0 uA	NA	G9,G10
	NA	± 107.0 uA	NA	G11
	NA	± 93.0 uA	NA	G12
	NA	± 58.5 uA	NA	G13
	NA	± 31.0 uA	NA	G14,G15
	NA	± 75.8 uA	NA	G16
	NA	± 31.0 uA	NA	G17
	NA	± 52.0 uA	NA	G18-G25
	NA	± 124.0 uA	NA	G26
	NA	± 84.0 uA	NA	G27,G28
	NA	± 168.0 uA	NA	G29
	NA	± 84.0 uA	NA	G30,G31
	NA	± 13.8 uA	NA	G32-G35
	NA	± 820.0 uA	NA	G36
	NA	± 52.0 uA	NA	G37-G44
	NA	± 107.0 uA	NA	G45
	NA	± 21.0 uA	NA	G46
	NA	± 47.0 uA	NA	G47-G54
	NA	± 58.0 uA	NA	G55-G62
	NA	± 31.0 uA	NA	G63,G64
	NA	± 124.0 uA	NA	G65
	NA	± 75.8 uA	NA	G66
	NA	± 62.0 uA	NA	G67,G68
	NA	± 31.0 uA	NA	G69
	NA	± 62.0 uA	NA	G70
	NA	± 93.0 uA	NA	G71,G72
	NA	± 62.0 uA	NA	G73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	PIPE BIT 7	2	-0.4	2.4	4.5	V	1-4
H2	PIPE BIT 6	3					
H3	PIPE BIT 5	4					
H4	PIPE BIT 4	5					
H5	PIPE BIT 3	6					
H6	PIPE BIT 2	7					
H7	PIPE BIT 1	8					
H8	PIPE BIT 0	9					
H9	BUS BIT 10	36					
H10	BUS BIT 0	52					
H11	BUS BIT 1	53					
H12	BUS BIT 2	54					
H13	BUS BIT 3	55					
H14	BUS BIT 4	56					
H15	BUS BIT 5	57					
H16	BUS BIT 6	58					
H17	BUS BIT 7	59					
H18	DATA BIT 12	63					
H19	DATA BIT 13	64	-0.4	2.4	4.5	V	1-4
H20	DATA BIT 14	65	-0.16	2.4	4.5	V	1-4
H21	DATA BIT 15	66	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	BOOT PRES	67	-0.16	2.4	4.5	V	1-4
H23	EMUL BIT 0	68	-0.4	2.4	4.5	V	1-4
H24	EMUL BIT 1	69					
H25	EMUL BIT 2	70					
H26	EMUL BIT 3	71					
H27	EMUL BIT 4	72					
H28	EMUL BIT 5	73					
H29	EMUL BIT 6	74					
H30	EMUL BIT 7	76					
H31	EMUL BIT 8	77					
H32	EMUL BIT 9	78					
H33	EMUL BIT 10	79					
H34	EMUL BIT 11	80					
H35	EMUL BIT 12	82					
H36	EMUL BIT 13	83					
H37	EMUL BIT 14	84					
H38	EMUL BIT 15	85					
H39	BUS BIT 8	95					
H40	BUS BIT 9	96					
H41	JMP COND T	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	3.13 A	H1-H41
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA	NA NA NA	H1-H19 H20-H22 H23-H41
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 NA NA	 H1-H41 H1-H41
Output voltage measurement	NA	± 105 mV	NA	H1-H41

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

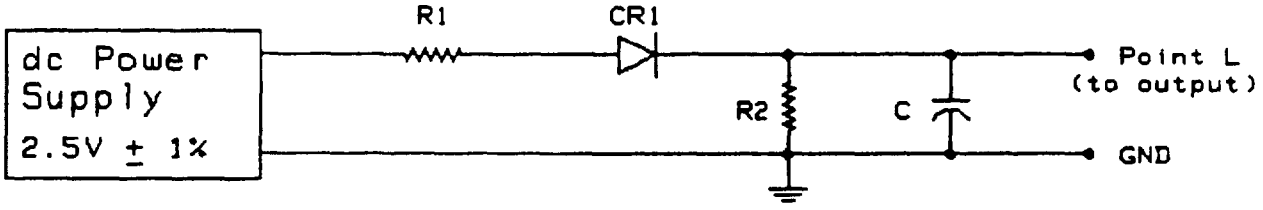
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9,36, 52,53,54,56,57,58, 59,63,64,68,69,70, 71,72,73,74,76,77, 78,79,80,82,83,84, 85,95,96,100	6200	6.0	0.5
65,66,67	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	PIPE BIT 7	2	6.0	0.0	0.5	V	1-4
J2	PIPE BIT 6	3					
J3	PIPE BIT 5	4					
J4	PIPE BIT 4	5					
J5	PIPE BIT 3	6					
J6	PIPE BIT 2	7					
J7	PIPE BIT 1	8					
J8	PIPE BIT 0	9					
J9	BUS BIT 10	36					
J10	BUS BIT 0	52					
J11	BUS BIT 1	53					
J12	BUS BIT 2	54					
J13	BUS BIT 3	55					
J14	BUS BIT 4	56					
J15	BUS BIT 5	57					
J16	BUS BIT 6	58					
J17	BUS BIT 7	59					
J18	DATA BIT 12	63					
J19	DATA BIT 13	64	6.0	0.0	0.5	V	1-4
J20	DATA BIT 14	65	1.6	0.0	0.5	V	1-4
J21	DATA BIT 15	66	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	BOOT PRES	67	1.6	0.0	0.5	V	1-4
J23	EMUL BIT 0	68	6.0	0.0	0.5	V	1-4
J24	EMUL BIT 1	69					
J25	EMUL BIT 2	70					
J26	EMUL BIT 3	71					
J27	EMUL BIT 4	72					
J28	EMUL BIT 5	73					
J29	EMUL BIT 6	74					
J30	EMUL BIT 7	76					
J31	EMUL BIT 8	77					
J32	EMUL BIT 9	78					
J33	EMUL BIT 10	79					
J34	EMUL BIT 11	80					
J35	EMUL BIT 12	82					
J36	EMUL BIT 13	83					
J37	EMUL BIT 14	84					
J38	EMUL BIT 15	85					
J39	BUS BIT 8	95					
J40	BUS BIT 9	96					
J41	JMP COND T	100	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	3.13 A	J1-J41
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA	NA NA NA	J1-J19 J20-J22 J23-J41
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	NA NA	J1-J41 J1-J41
Output voltage measurement	NA	± 25 mV	NA	J1-J41

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	BUS BIT 10	36	2.4	-0.05	0.05	mA	1-3
K2	BUS BIT 0	52					
K3	BUS BIT 1	53					
K4	BUS BIT 2	54					
K5	BUS BIT 3	55					
K6	BUS BIT 4	56					
K7	BUS BIT 5	57					
K8	BUS BIT 6	58					
K9	BUS BIT 7	59					
K10	BUS BIT 8	95					
K11	BUS BIT 9	96					
K12	JMP COND T	100	2.4	-0.05	0.05	mA	1-3
Low level							
K13	BUS BIT 10	36	0.4	-0.05	0.05	mA	1-3
K14	BUS BIT 0	52					
K15	BUS BIT 1	53					
K16	BUS BIT 2	54					
K17	BUS BIT 3	55					
K18	BUS BIT 4	56					
K19	BUS BIT 5	57	0.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K20	BUS BIT 6	58	0.4	-0.05	0.05	mA	1-3
K21	BUS BIT 7	59					
K22	BUS BIT 8	95					
K23	BUS BIT 9	96					
K24	JMP COND T	100	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	3.13 A	K1-K24
Forcing voltage, (Voh)	2.4 V	± 50 mV	NA	K1-K12
(Vol)	0.5 V	± 50 mV	NA	K13-K24
Pattern voltage high, (VH)	3.0 V	± 100 mV	NA	K1-K24
low, (VL)	0.4 V	± 100 mV	NA	K1-K24
Output current measurement	NA	± 5 uA	NA	K1-K24

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CLK Y REG	11	-18.0	-1.5	0.0	V	1-3
L2	CLK MDRI	12	-18.0	-1.5	0.0	V	1-3
L3	INTR PEND	13	-18.0	-1.2	0.0	V	1-3
L4	CLK PIPE	14					
L5	IR BIT 15	15					
L6	IR BIT 16	16					
L7	IR BIT 17	17					
L8	CTRL FRMT	18					
L9	PWR FAULT	19					
L10	DAT BIT 3	20					
L11	DAT BIT 2	21					
L12	DAT BIT 1	22					
L13	DAT BIT 0	23					
L14	DAT BIT 7	24					
L15	DAT BIT 6	26					
L16	DAT BIT 5	27					
L17	DAT BIT 4	28					
L18	CB/I STACK	29	-18.0	-1.2	0.0	V	1-3
L19	STACK R1	30	-18.0	-1.5	0.0	V	1-3
L20	STACK R0	32					
L21	CLK STACK	33	-18.0	-1.5	0.0	V	1-3

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Check the source to verify that this is the current version before use.

TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	STACK W1	34	-18.0	-1.5	0.0	V	1-3
L23	STACK W0	35					
L24	EC1 DEST	37					
L25	IR BIT 26	38					
L26	IR BIT 25	39					
L27	IR BIT 24	41	-18.0	-1.5	0.0	V	1-3
L28	PH N CLK	42	-18.0	-1.2	0.0	V	1-3
L29	DAT BIT 8	43					
L30	DAT BIT 9	44					
L31	DAT BIT 10	45					
L32	DAT BIT 11	46					
L33	DAT BIT 12	47					
L34	DAT BIT 13	48					
L35	DAT BIT 14	49					
L36	DAT BIT 15	50					
L37	J MAP OE	61	-18.0	-1.2	0.0	V	1-3
L38	STOP KEY 2	62	-18.0	-1.5	0.0	V	1-3
L39	STOP KEY 1	86					
L40	BT 1/ BT 2	87	-18.0	-1.5	0.0	V	1-3
L41	EN SB 18-1B	88	-18.0	-1.2	0.0	V	1-3
L42	IR BIT 14	89	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L43	ALU NEG	91	-18.0	-1.2	0.0	V	1-3
L44	ALU ZERO	92					
L45	ALU CARRY	93					
L46	ALU OVERFL	94					
L47	IR BIT 12	97					
L48	IR BIT 13	98					
L49	IR BIT 41	99	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	Off	NA	NA	L1-L49
Forcing current, (Iin)	-18 mA	± 1.8 mA	NA	L1-L49
Pattern voltage high, (VH)	NA	NA	NA	L1-L49
low, (VL)	NA	NA	NA	L1-L49

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Input voltage measurement	NA	± 75 mV	NA	L1,L2
	NA	± 60 mV	NA	L3-L18
	NA	± 75 mV	NA	L19-L27
	NA	± 60 mV	NA	L28-L37
	NA	± 75 mV	NA	L38-L40
	NA	± 60 mV	NA	L41-L49

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.2	6.1	uF	1-4
M2	PSFC	51	2.6	4.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	NA	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	NA	M1,M2
Capacitance measurement	NA NA	± 145 nF ± 115 nF	NA NA	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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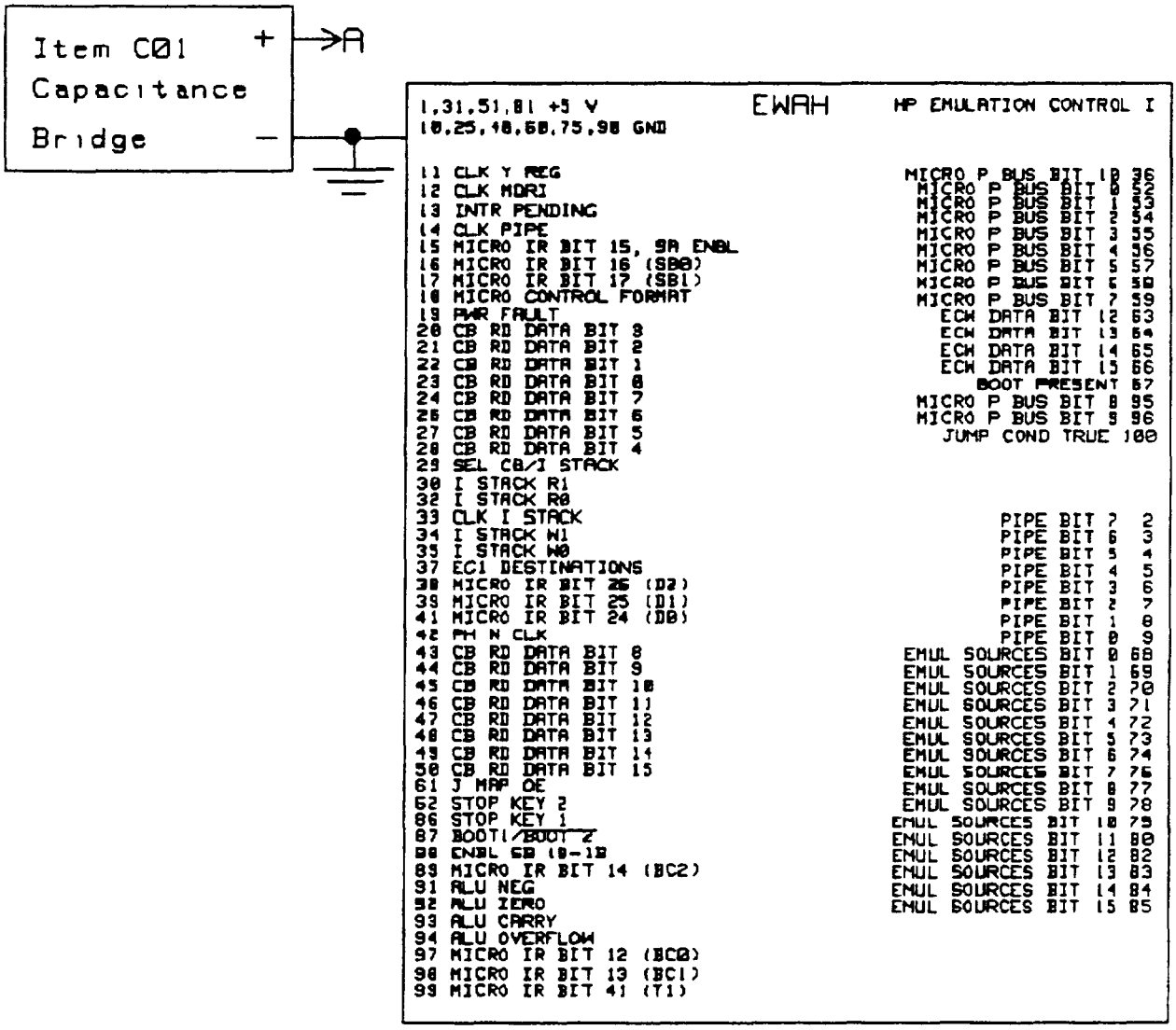


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	870	2088	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	3.13 A	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	NA	N
low, (VL)	0.0 V	± 100 mV	NA	N
Power supply current measurement	NA	± 60.9 mA	NA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (325 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (325 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	3.13 A 3.13 A	P1 P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 NA NA	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 NA NA NA	 P1 P2 P1,P2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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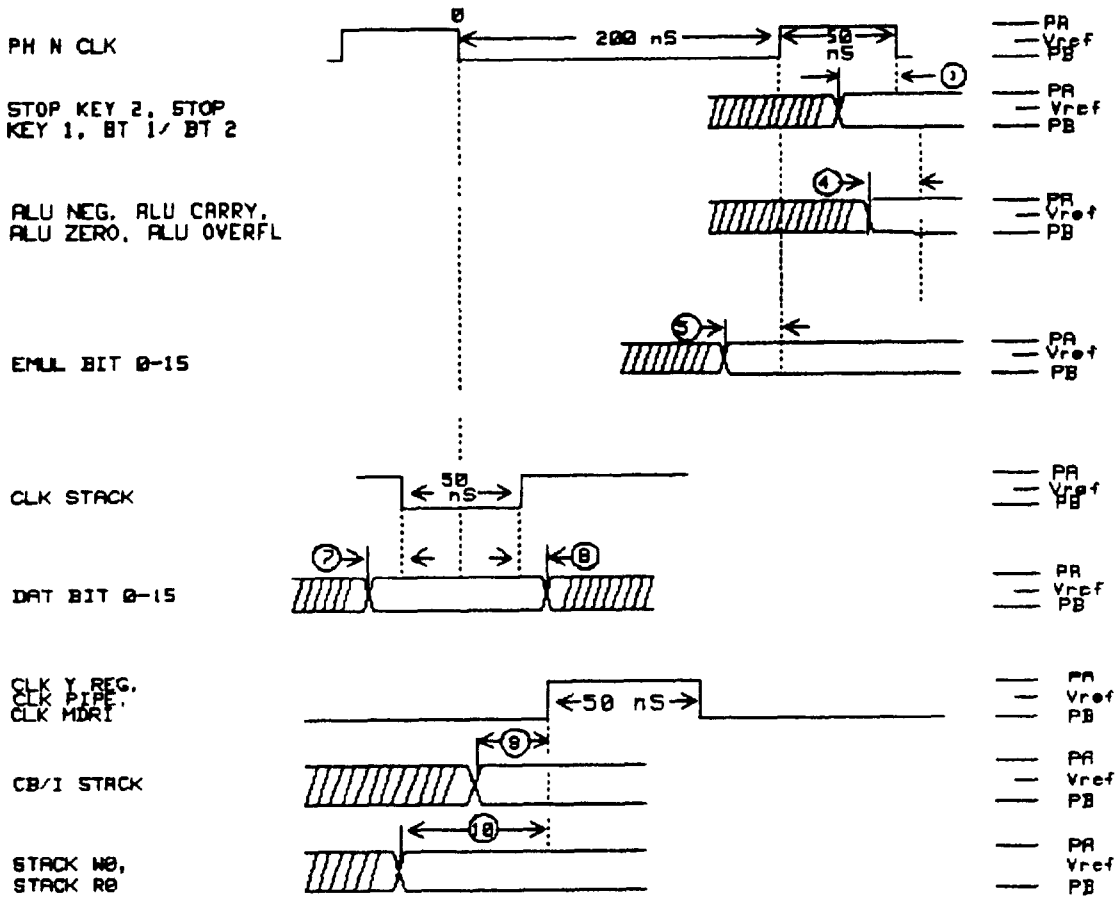


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) STOP KEY 2, STOP KEY 1 and BT 1/ BT 2 to PH N CLK. Setup time: minimum 30 nS.

2. (4) ALU NEG, ALU CARRY, ALU ZERO, and ALU OVERFL to PH N CLK. Setup time: minimum 15 nS.

3. (5) EMUL BIT 0-15 to PH N CLK. Setup time: minimum 10 nS.

4. (7) DAT BIT 0-15 to CLK STACK. Setup time: minimum 15 nS.

5. (8) CLK STACK to DAT BIT 0-15. Hold time: minimum 15 nS.

6. (9) CB/I STACK to CLK PIPE, CLK Y REG, and CLK MDRI. Setup time: minimum 40 nS.

7. (10) STACK W0 and STACK R0 to CLK Y REG, CLR PIPE, and CLK MDRI. Setup time: minimum 85 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (325 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (325 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (325 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (325 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	3.13 A 3.13 A	Q1,Q3 Q2,Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 NA NA	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Output voltage verification				
high, (Voh)	NA	± 100 mV	NA	Q1,Q3
high, (Voh)	NA	± 100 mV	NA	Q2,Q4
low, (Vol)	NA	± 100 mV	NA	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

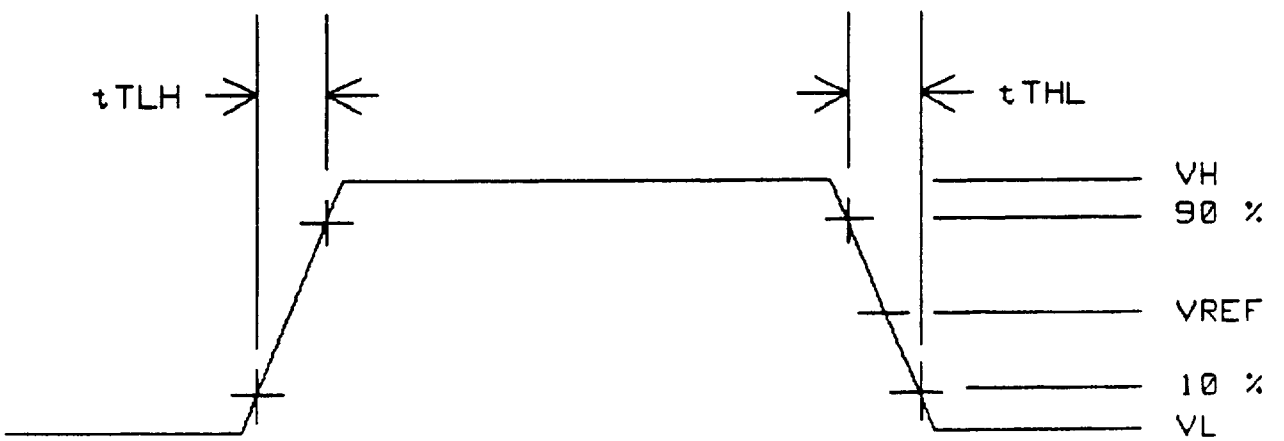
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	STOP KEY 2 to PH N CLK	30	30	30	--	--	--	nS	12
1	tsu	STOP KEY 1 to PH N CLK	30	30	30	--	--	--	nS	12
1	tsu	BT 1/ BT 2 to PH N CLK	30	30	30	--	--	--	nS	12
2	tpd	PH N CLK to JMP COND T	--	--	--	90	90	90	nS	12
3	tpd	PH N CLK to ECW BIT 12	--	--	--	140	140	140	nS	12
3	tpd	PH N CLK to ECW BIT 13	--	--	--	140	140	140	nS	12
3	tpd	PH N CLK to ECW BIT 14	--	--	--	140	140	140	nS	12
3	tpd	PH N CLK to ECW BIT 15	--	--	--	140	140	140	nS	12
4	tsu	ALU NEG to PH N CLK	15	15	15	--	--	--	nS	12
4	tsu	ALU CARRY to PH N CLK	15	15	15	--	--	--	nS	12
4	tsu	ALU ZERO to PH N CLK	15	15	15	--	--	--	nS	12
4	tsu	ALU OVER-FL to PH N CLK	15	15	15	--	--	--	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	EMUL BIT 0 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 1 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 2 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 3 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 4 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 5 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 6 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 7 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 8 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 9 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 10 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 11 to PH N CLK	10	10	10	--	--	--	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	EMUL BIT 12 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 13 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 14 to PH N CLK	10	10	10	--	--	--	nS	12
5	tsu	EMUL BIT 15 to PH N CLK	10	10	10	--	--	--	nS	12
6	tpd	J MAP OE to BUS BIT 0	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 1	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 2	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 3	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 4	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 5	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 6	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 7	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 8	--	--	--	40	40	40	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	J MAP OE to BUS BIT 9	--	--	--	40	40	40	nS	12
6	tpd	J MAP OE to BUS BIT 10	--	--	--	40	40	40	nS	12
7	tsu	DAT BIT 0 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 1 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 2 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 3 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 4 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 5 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 6 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 7 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 8 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 9 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 10 to CLK STACK	15	15	15	--	--	--	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	DAT BIT 11 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 12 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 13 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 14 to CLK STACK	15	15	15	--	--	--	nS	12
7	tsu	DAT BIT 15 to CLK STACK	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 0	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 1	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 2	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 3	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 4	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 5	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 6	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 7	15	15	15	--	--	--	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	CLK STACK to DAT BIT 8	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 9	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 10	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 11	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 12	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 13	15	15	15	--	--	--	nS	12
8	th	CLK STACK to DAT BIT 14	15	15	15	--	--	--	nS	12
8	tsu	CLK STACK to DAT BIT 15	15	15	15	--	--	--	nS	12
9	tsu	CB/I STACK to CLK PIPE	40	40	40	--	--	--	nS	12
9	tsu	CB/I STACK to CLK Y REG	40	40	40	--	--	--	nS	12
9	tsu	CB/I STACK to CLK MDRI	40	40	40	--	--	--	nS	12
10	tsu	STACK W0 to CLK Y REG	85	85	85	--	--	--	nS	12
10	tsu	STACK W0 to CLK PIPE	85	85	85	--	--	--	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tsu	STACK W0 to CLK MDRI	85	85	85	--	--	--	nS	12
10	tsu	STACK R0 to CLK Y REG	85	85	85	--	--	--	nS	12
10	tsu	STACK R0 to CLK PIPE	85	85	85	--	--	--	nS	12
10	tsu	STACK R0 to CLK MDRI	85	85	85	--	--	--	nS	12
11	tpd	PH N CLK to PIPE BIT 0	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 1	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 2	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 3	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 4	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 5	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 6	--	--	--	50	50	50	nS	12
11	tpd	PH N CLK to PIPE BIT 7	--	--	--	50	50	50	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 0	--	--	--	25	25	25	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	EN SB 18-1B to EMUL BIT 1	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 2	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 3	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 4	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 5	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 6	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 7	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 8	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 9	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 10	--	--	--	25	25	25	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	EN SB 18-1B to EMUL BIT 11	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 12	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 13	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 14	--	--	--	25	25	25	nS	12
12	tpd	EN SB 18-1B to EMUL BIT 15	--	--	--	25	25	25	nS	12

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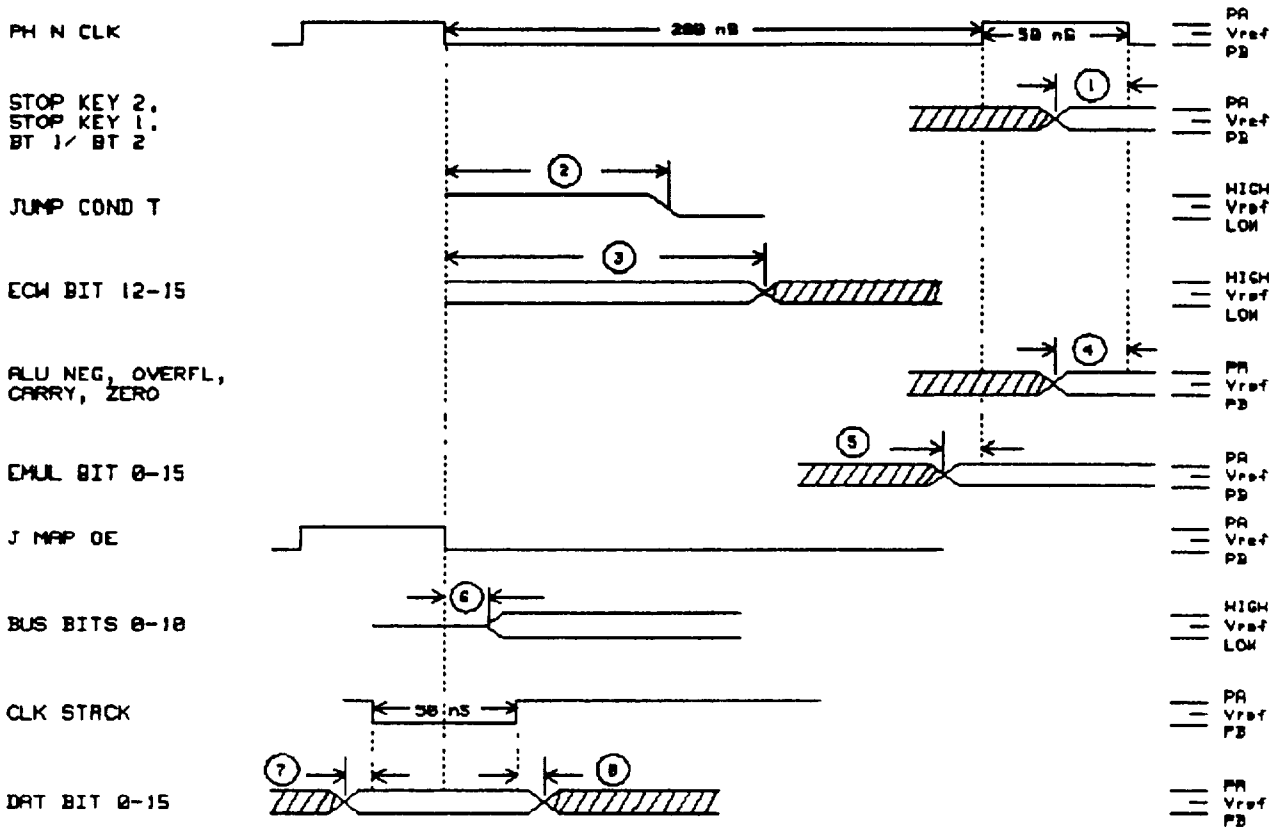


FIGURE 12a. Dynamic waveforms for table XVII.

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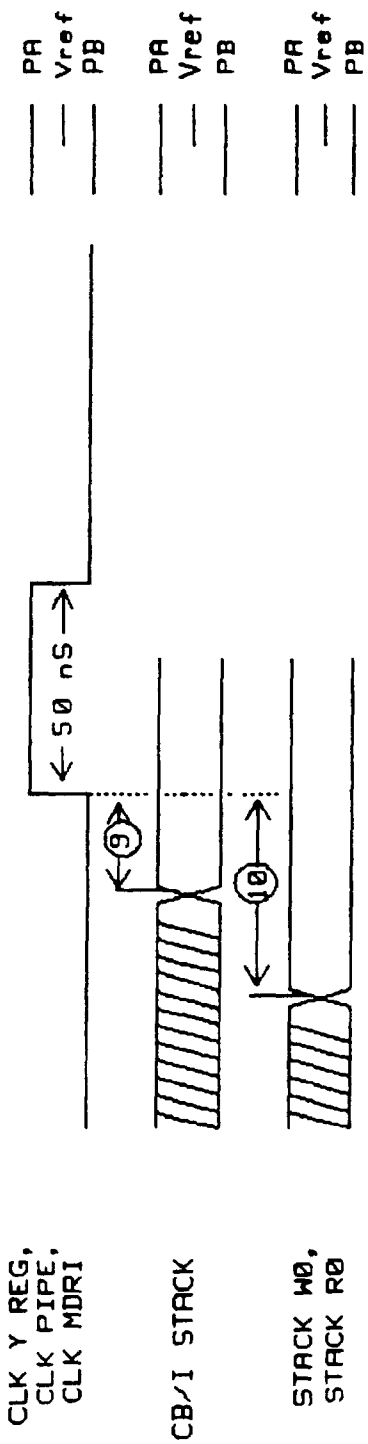


FIGURE 12b. Dynamic waveforms for table XVII.

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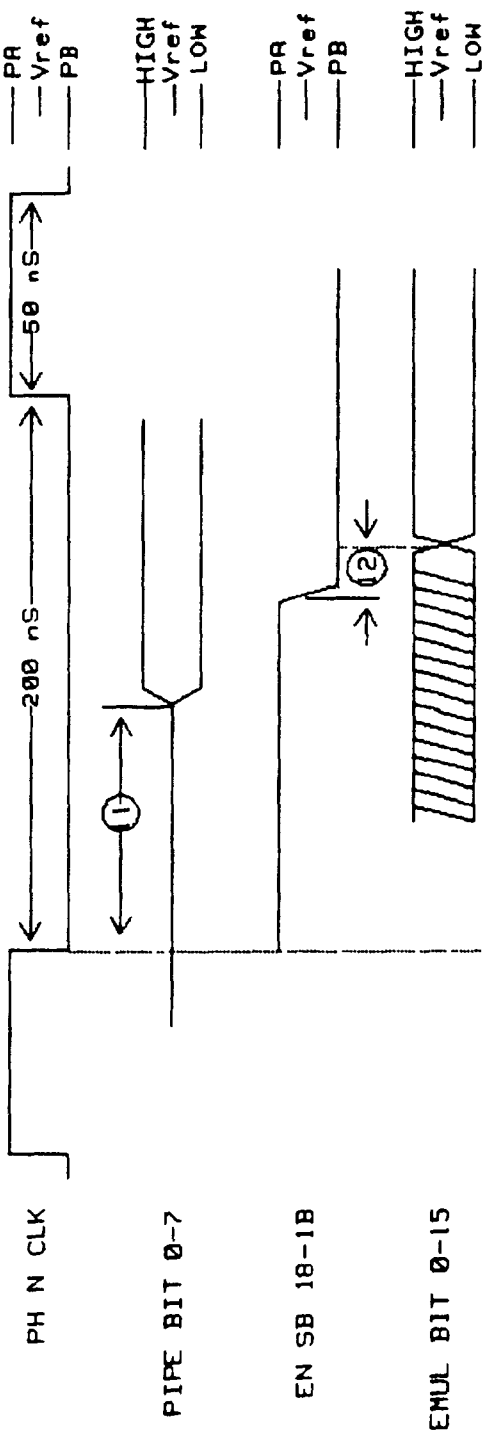


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CLK Y REG	11	4	100	pF	1-4
R2	CLK MDRI	12	4	100	pF	1-4
R3	INTR PEND	13	4	150	pF	1-4
R4	CLK PIPE	14	4	150	pF	1-4
R5	IR BIT 15	15	4	125	pF	1-4
R6	IR BIT 16	16	2	50	pF	1-4
R7	IR BIT 17	17	2	50	pF	1-4
R8	CTRL FRMT	18	6	125	pF	1-4
R9	PWR FAULT	19	2	100	pF	1-4
R10	DAT BIT 3	20	4	75	pF	1-4
R11	DAT BIT 2	21				
R12	DAT BIT 1	22				
R13	DAT BIT 0	23				
R14	DAT BIT 7	24				
R15	DAT BIT 6	26				
R16	DAT BIT 5	27				
R17	DAT BIT 4	28	4	75	pF	1-4
R18	CB/I STACK	29	8	125	pF	1-4
R19	STACK R1	30	8	100	pF	1-4
R20	STACK R0	32	8	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CLK STACK	33	8	125	pF	1-4
R22	STACK W1	34	8	100	pF	1-4
R23	STACK W0	35	8	100	pF	1-4
R24	EC1 DEST	37	2	50	pF	1-4
R25	IR BIT 26	38				
R26	IR BIT 25	39				
R27	IR BIT 24	41	2	50	pF	1-4
R28	PH N CLK	42	16	125	pF	1-4
R29	DAT BIT 8	43	4	75	pF	1-4
R30	DAT BIT 9	44	4	75	pF	1-4
R31	DAT BIT 10	45	4	100	pF	1-4
R32	DAT BIT 11	46	4	100	pF	1-4
R33	DAT BIT 12	47	4	75	pF	1-4
R34	DAT BIT 13	48				
R35	DAT BIT 14	49	4	75	pF	1-4
R36	DAT BIT 15	50	4	100	pF	1-4
R37	J MAP OE	61	2	175	pF	1-4
R38	STOP KEY 2	62	2	100	pF	1-4
R39	STOP KEY 1	86	2	75	pF	1-4
R40	BT 1/ BT 2	87	2	125	pF	1-4
R41	EN SB 18-1B	88	8	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R42	IR BIT 14	89	2	100	pF	1-4
R43	ALU NEG	91	4	75	pF	1-4
R44	ALU ZERO	92	4	75	pF	1-4
R45	ALU CARRY	93	2	50	pF	1-4
R46	ALU OVERFL	94	4	75	pF	1-4
R47	IR BIT 12	97	6	75	pF	1-4
R48	IR BIT 13	98	6	100	pF	1-4
R49	IR BIT 41	99	4	75	pF	1-4
Bidirectionals						
R50	PIPE BIT 7	2	4	75	pF	1-4
R51	PIPE BIT 6	3				
R52	PIPE BIT 5	4				
R53	PIPE BIT 4	5	4	75	pF	1-4
R54	PIPE BIT 3	6	4	100	pF	1-4
R55	PIPE BIT 2	7				
R56	PIPE BIT 1	8				
R57	PIPE BIT 0	9	4	100	pF	1-4
Outputs						
R58	BUS BIT 10	36	2	75	pF	1-4
R59	BUS BIT 0	52	2	50	pF	1-4
R60	BUS BIT 1	53	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	BUS BIT 2	54	2	50	pF	1-4
R62	BUS BIT 3	55				
R63	BUS BIT 4	56				
R64	BUS BIT 5	57	2	50	pF	1-4
R65	BUS BIT 6	58	2	75	pF	1-4
R66	BUS BIT 7	59	2	50	pF	1-4
R67	DAT BIT 12	63	4	100	pF	1-4
R68	DAT BIT 13	64	4	125	pF	1-4
R69	DAT BIT 14	65				
R70	BOOT PRES	66	4	125	pF	1-4
R71	DAT BIT 15	67	2	75	pF	1-4
R72	BUS BIT 8	95	2	50	pF	1-4
R73	BUS BIT 9	96				
R74	JMP COND T	100	2	50	pF	1-4
Bidirectionals						
R75	EMUL BIT 0	68	6	100	pF	1-4
R76	EMUL BIT 1	69	6	100	pF	1-4
R77	EMUL BIT 2	70	6	75	pF	1-4
R78	EMUL BIT 3	71	6	100	pF	1-4
R79	EMUL BIT 4	72				
R80	EMUL BIT 5	73	6	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R81	EMUL BIT 6	74	6	100	pF	1-4
R82	EMUL BIT 7	76	6	100	pF	1-4
R83	EMUL BIT 8	77	2	150	pF	1-4
R84	EMUL BIT 9	78	2	175	pF	1-4
R85	EMUL BIT 10	79				
R86	EMUL BIT 11	80	2	175	pF	1-4
R87	EMUL BIT 12	82	2	150	pF	1-4
R88	EMUL BIT 13	83	2	150	pF	1-4
R89	EMUL BIT 14	84	2	175	pF	1-4
R90	EMUL BIT 15	85	2	175	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	NA	R1-R90
Capacitance meter voltage	15 mV	± 1.5 mV	NA	R1-R90
Capacitance	NA	± 4.8 pF	NA	R1,R2
	NA	± 7.3 pF	NA	R3,R4
	NA	± 6.05 pF	NA	R5
	NA	± 2.4 pF	NA	R6,R7
	NA	± 5.95 pF	NA	R8

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Capacitance measurement	NA	± 4.9 pF	NA	R9
	NA	± 3.55 pF	NA	R10-R17
	NA	± 5.85 pF	NA	R18
	NA	± 4.6 pF	NA	R19,R20
	NA	± 5.85 pF	NA	R21
	NA	± 4.6 pF	NA	R22,R23
	NA	± 2.4 pF	NA	R24-R27
	NA	± 5.45 pF	NA	R28
	NA	± 3.55 pF	NA	R29,R30
	NA	± 4.8 pF	NA	R31,R32
	NA	± 3.55 pF	NA	R33-R35
	NA	± 4.8 pF	NA	R36
	NA	± 8.65 pF	NA	R37
	NA	± 4.9 pF	NA	R38
	NA	± 3.65 pF	NA	R39
	NA	± 6.15 pF	NA	R40
	NA	± 5.85 pF	NA	R41
	NA	± 4.9 pF	NA	R42
	NA	± 3.55 pF	NA	R43,R44
	NA	± 2.4 pF	NA	R45
	NA	± 3.55 pF	NA	R46
	NA	± 3.45 pF	NA	R47
	NA	± 4.7 pF	NA	R48
	NA	± 3.55 pF	NA	R49-R53
	NA	± 4.8 pF	NA	R54-R57
	NA	± 3.65 pF	NA	R58
	NA	± 2.4 pF	NA	R59-R64
	NA	± 3.65 pF	NA	R65
	NA	± 2.4 pF	NA	R66
	NA	± 4.8 pF	NA	R67
	NA	± 6.05 pF	NA	R68-R70
	NA	± 3.65 pF	NA	R71
	NA	± 2.4 pF	NA	R72-R74
	NA	± 4.7 pF	NA	R75,R76
	NA	± 3.45 pF	NA	R77
	NA	± 4.7 pF	NA	R78-R82
	NA	± 7.4 pF	NA	R83
	NA	± 8.65 pF	NA	R84-R86

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Capacitance measurement	NA NA	± 7.4 pF ± 8.65 pF	NA NA	R87,R88 R89,R90

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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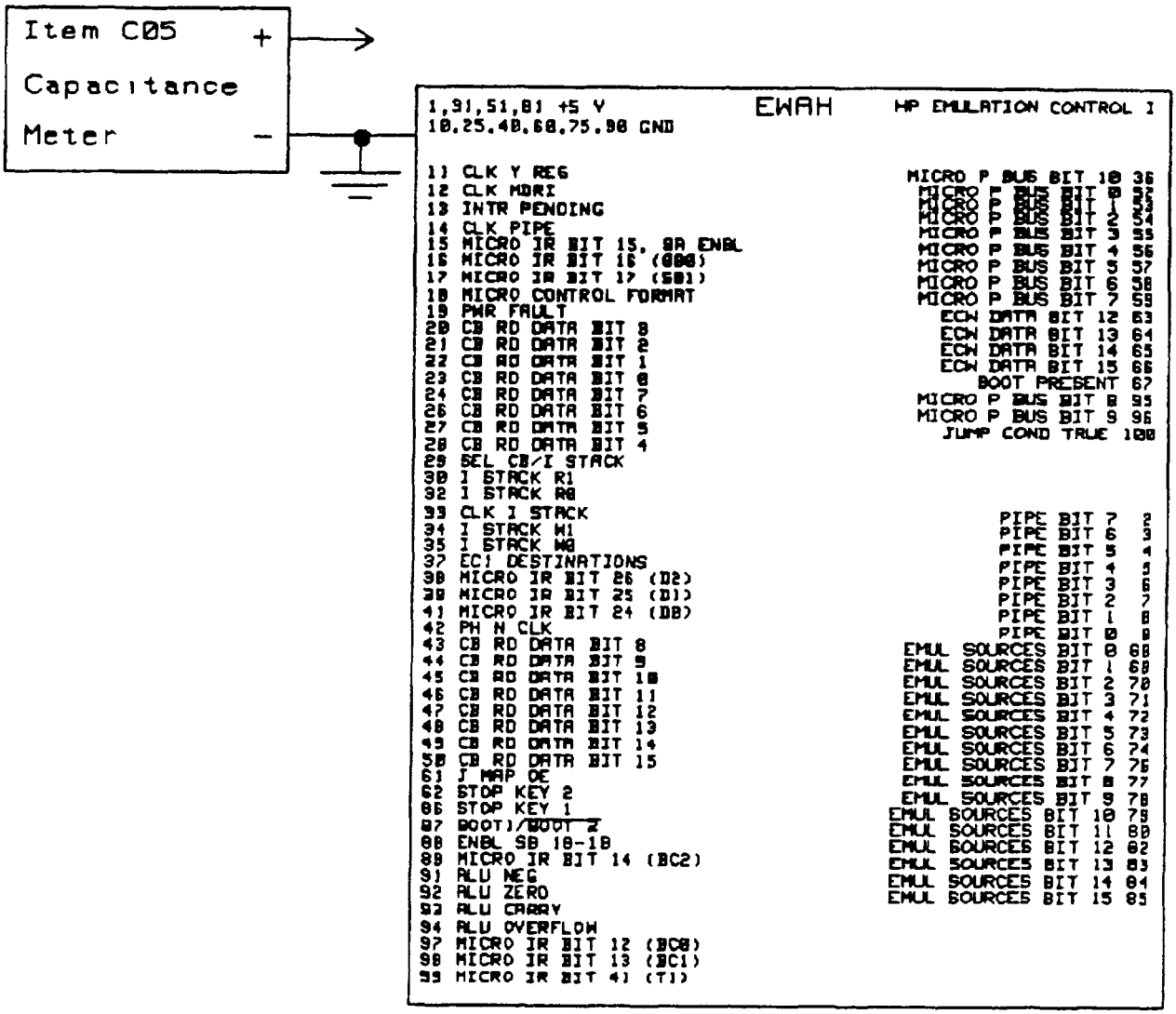


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latch up test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S	CLK PIPE	14	-18.0	-1.2	2.4	V	1-3

NOTES:

1. Input clamp diode latch up voltage test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	3.13 A	S
Forcing current, (Iin)	-18 mA	± 1.8 mA	NA	S
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	NA NA	S S
Input voltage measurement	NA	± 180 mV	NA	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

(d) Pulse duration shall be 500 ± 20 nS.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	3.13 A	T1,T2
Pattern voltage				
high, (VH)	3.0 V	± 100 mV	NA	T1,T2
low, (VL)	0.0 V	± 100 mV	NA	T1,T2
PSTA measurement	NA	± 20 mV	NA	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

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NOTES: - Continued.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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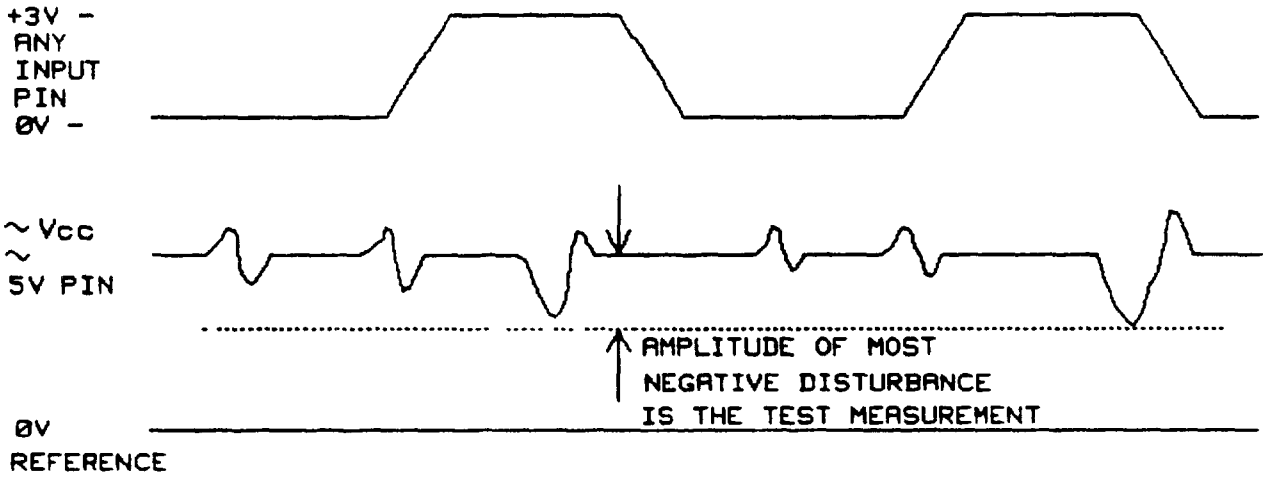
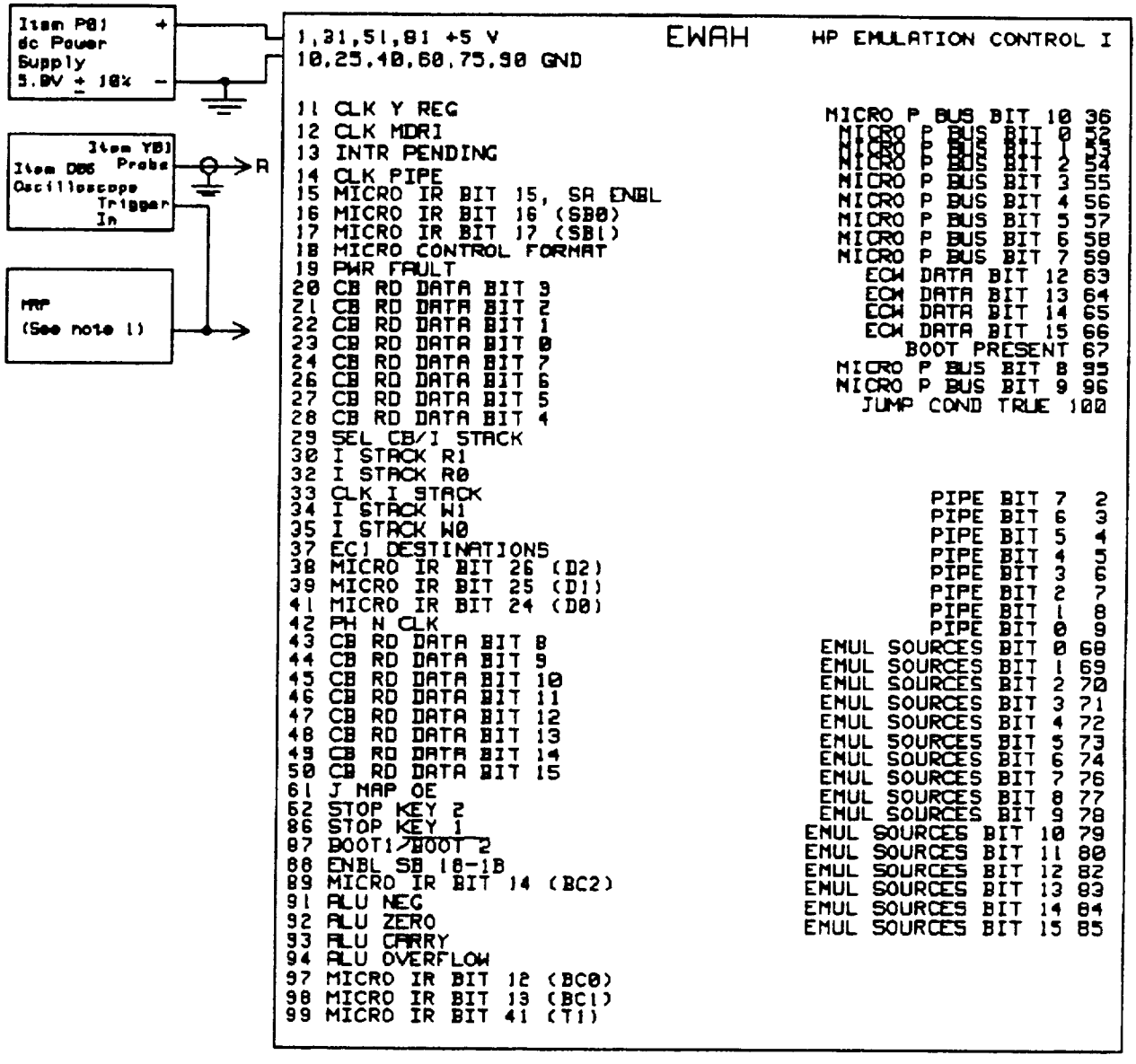


FIGURE 14. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

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